

Service Manual

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SAFETY CONSIDERATIONS

Installing, starting up, and servicing air-conditioning equipment can be hazardous due to unit pressures, electrical components, and equipment location (roofs, elevated structures, etc.).

Only trained, qualified installers and service mechanics should install, start-up, and service this equipment.

Untrained personnel can perform basic maintenance functions such as coil cleaning. All other operations should be performed by trained service personnel **only**.

When working on the equipment, observe the precautions in the literature and on tags, stickers, and labels attached to the equipment.

Follow all safety codes. Wear safety glasses and work gloves. Keep a quenching cloth and fire extinguisher nearby when brazing. Use care in handling, rigging, and setting bulky equipment.

Read this manual thoroughly and follow all the warnings or cautions included in the literature and attached to the unit. Consult local building codes and the National Electrical Code (NEC) for special requirements.

Recognize safety information. This is the safety-alert symbol  . When you see this symbol on the unit and in instructions or manuals, be alert to the potential for personal injury. Understand these signal words: **DANGER**, **WARNING**, and **CAUTION**.

These words are used with the safety-alert symbol. **DANGER** identifies the most serious of hazards which will result in severe personal injury or death. **WARNING** signifies hazards which could result in personal injury or death. **CAUTION** is used to identify unsafe practices which may result in minor personal injury or product and property damage. **NOTE** is used to highlight suggestions which will result in enhanced installation, reliability, or operation.



WARNING

ELECTRICAL SHOCK HAZARD

Failure to follow this warning could result in personal injury or death. Before installing, modifying, or servicing the unit, the main electrical disconnect switch must be in the **OFF** position. There may be more than 1 disconnect switch. Lock out and tag switch with a suitable warning label.



WARNING



EXPLOSION HAZARD

Failure to follow this warning could result in death, serious personal injury, and/or property damage.

Never use air or gases containing oxygen for leak testing or operating refrigerant compressors. Pressurized mixtures of air or gases containing oxygen can lead to an explosion.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. **DO NOT** bury more than 36 in. (914 mm) of refrigerant pipe in the ground. If any section of pipe is buried, there must be a 6 in. (152 mm) vertical rise to the valve connections on the outdoor units. If more than the recommended length is buried, refrigerant may migrate to the cooler buried section during the extended periods of unit shutdown. This causes refrigerant slugging and could possibly damage the compressor at start-up.

INTRODUCTION

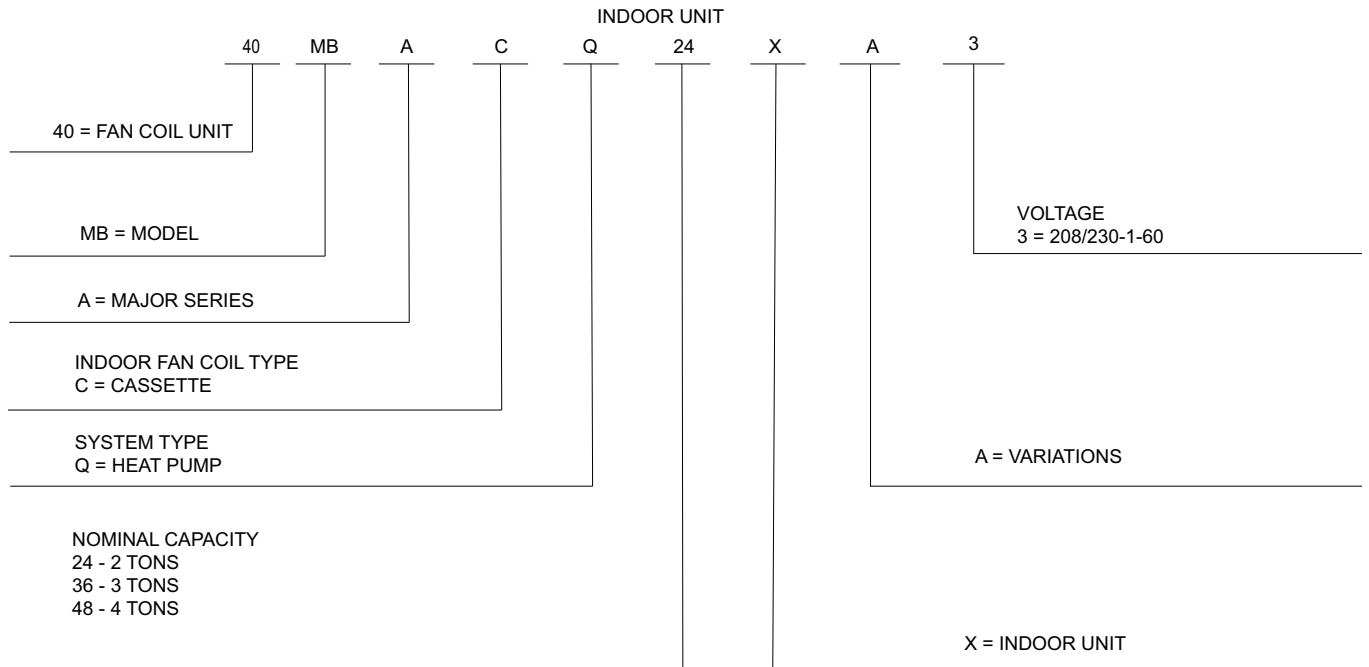
This service manual provides the necessary information to service, repair, and maintain the indoor units. This manual has an appendix with data required to perform troubleshooting. Use the Table of Contents to locate a desired topic.

INDOOR UNIT SIZES

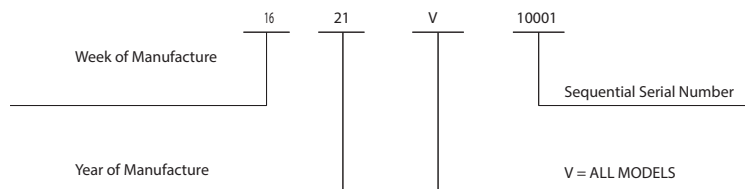
Table 1 — Indoor Unit Sizes

kBTUh	V-Ph-Hz	Model No.
24	208/230-60	40MBCAQ24---3
36		40MBCAQ36---3
48		40MBCAQ48---3

MODEL NUMBER NOMENCLATURE



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Use of the AHRI Certified TM Mark indicates a manufacturer's participation in the program. For verification of certification for individual products, go to www.ahridirectory.org.



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ELECTRICAL DATA

Table 2 — Electrical Data

UNIT SIZE	OPER. VOLTAGE MAX/MIN	INDOOR FAN				MAX FUSE CB AMP
		V-PH-HZ	FLA	HP	W	
24	253/187	208-230/1/60	1	0.06	106	Refer to the outdoor unit installation instructions - Indoor unit powered by the outdoor unit
36			1	0.168	240	
48			1	0.168		

*Permissible limits of the voltage range at which the unit will operate satisfactorily.

LEGEND

FLA - Full Load Amps

WIRING

All wires must be sized per NEC (National Electrical Code) or CEC (Canadian Electrical Code) and local codes. Use Electrical Data table MCA (minimum circuit amps) and MOC (maximum over current protection) to correctly size the wires and the disconnect fuse or breakers respectively.

Recommended Connection Method for Power and Communication Wiring:

Power and Communication Wiring: (24K)

The main power is supplied to the outdoor unit. The field supplied 14/3 power/communication wiring from the outdoor unit to the indoor unit consists of four (4) wires and provides the power for the indoor unit. Two wires are high voltage AC power, one is communication wiring and the other is a ground wire.

To minimize communication interference: If installed in a high Electromagnetic field (EMF) area and communication issues exist, a 14/2 stranded shielded wire can be used to replace L2 and (S) between outdoor unit and indoor unit - landing the shield onto ground in the outdoor unit only.

Recommended Connection Method for Power and Communication Wiring (36K through 48K) Power Wiring:

The main power is supplied to the outdoor unit. The field supplied power wiring from the outdoor unit to the indoor unit consists of three (3) wires and provides the power for the indoor unit. Two wires are high voltage AC power and one is a ground wire. To minimize voltage drop, the factory recommended wire size is 14/2 stranded with a ground.

Communication Wiring:

A separate shielded stranded copper conductor only, with a 600 volt rating and double insulated copper wire, must be used as the communication wire from the outdoor unit to the indoor unit. Please use a separate shielded 16GA stranded control wire.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Wires should be sized based on NEC and local codes.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Be sure to comply with local codes while running wire from indoor unit to outdoor unit.

Every wire must be connected firmly. Loose wiring may cause terminal to overheat or result in unit malfunction. A fire hazard may also exist.

Ensure all wiring is tightly connected.

No wire should touch refrigerant tubing compressor or any moving parts.

Disconnecting means must be provided and shall be located within sight and readily accessible from the unit.

Connecting cable with conduit shall be routed through hole in the conduit panel.

CONNECTION DIAGRAMS

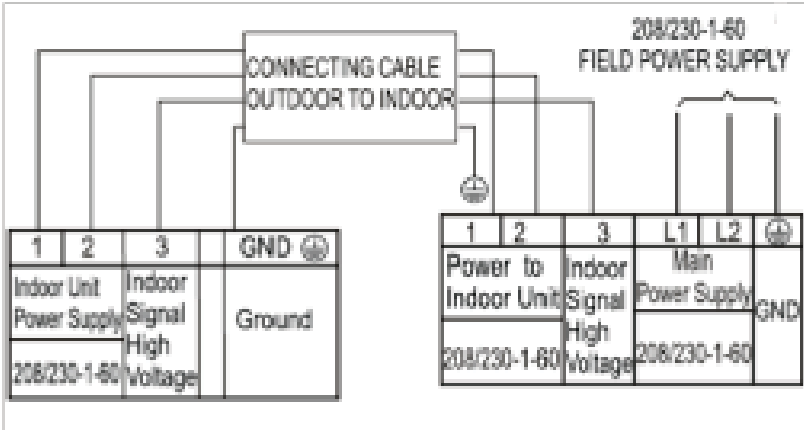


Fig. 1 — Connection Diagram 24K

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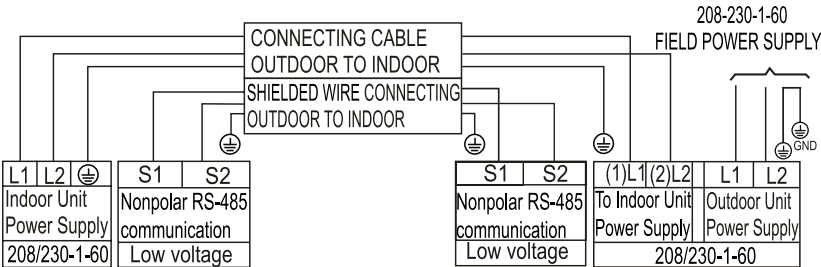


Fig. 2 — Connection Diagram 36K

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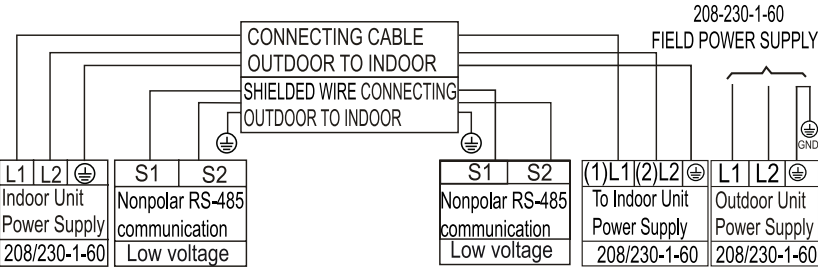


Fig. 3 — Connection Diagram 48K

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NOTES:

- 1. Do not use a thermostat wire for any connection between the indoor and outdoor units.
- 2. Unless otherwise stated, all connections between the indoor and outdoor units must adhere to the connections shown. The connections are polarity sensitive and improper wiring will result in a fault code.

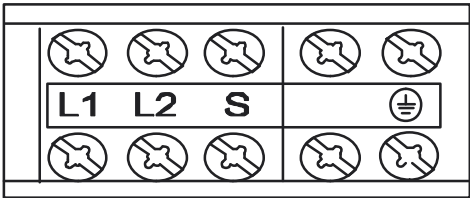


Fig. 4 —Control and Power Wiring on Indoor Unit

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WIRING DIAGRAMS

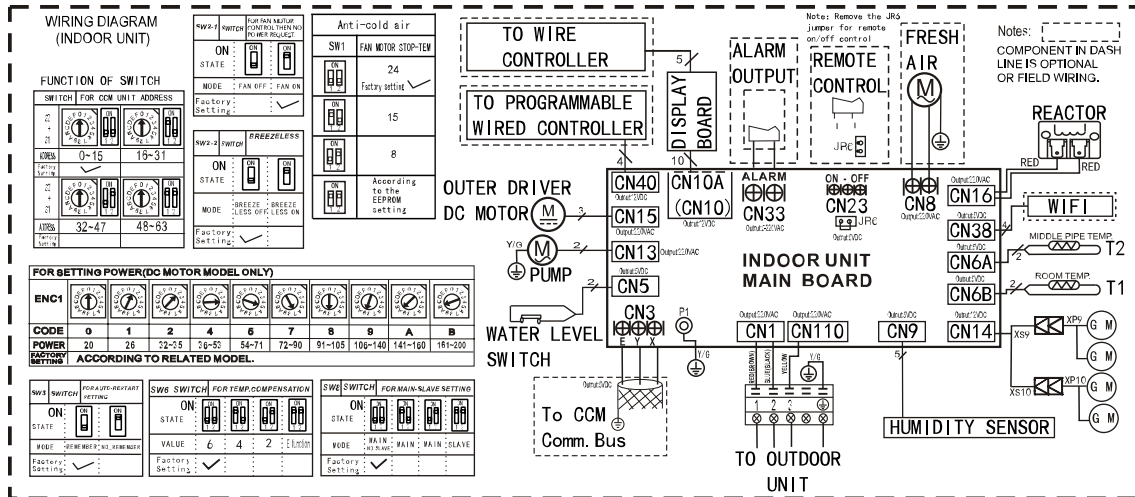


Fig. 5 — Wiring Diagram - Size 24K

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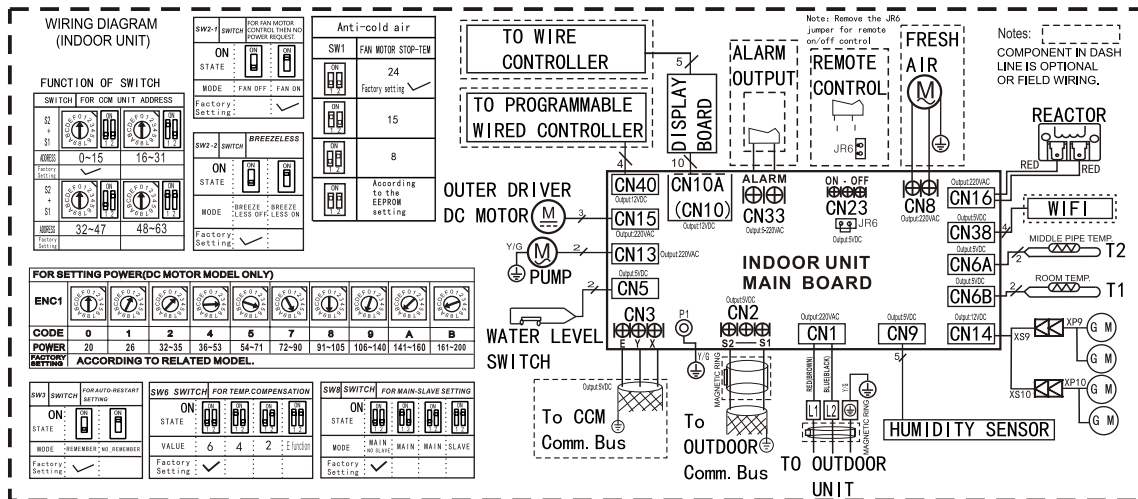


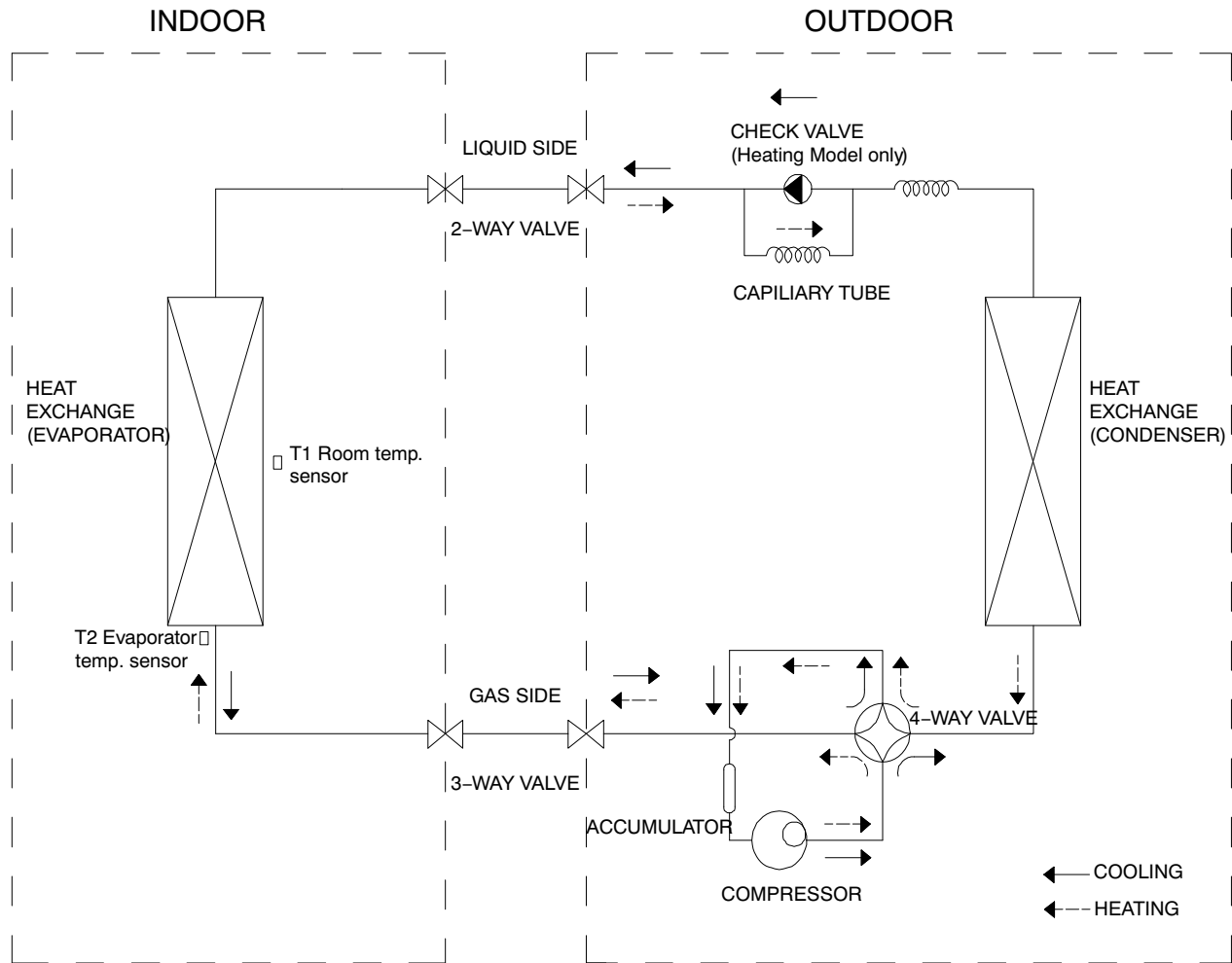
Fig. 6 — Wiring Diagram - Sizes 36K - 48K

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Table 3 — Wiring Diagram - Sizes 24K - 48K

INDOOR UNIT	
CODE	PART NAME
CN1	Input: 230VAC High voltage Connection of the terminal
CN3	Output: 0-5VDC Connection of the CCM
P1	Output: 0V Connection of the earth
CN5	Output: 1-5VDC Connection of the Water level switch
CN6	Output: 5VDC Connection of the Room and Pipe temperature
CN10A	Output: 12VDC Connection of the Display board
CN13	Output: 220VAC High voltage Connection of the Pump
CN14	Output: 12VDC Connection of the Swing motor
CN15	Output: 320VDC High voltage Connection of the DC Fan
CN16	Output: 320VDC High voltage Connection of the Reactor
CN23	Output: 1-12VDC Connection of the Remote switch
CN33	Output: 0V Connection of the Alarm
CN40	Output: 12VDC Connection of the Wire controller
CN110	Output: 24VDC between Pin2 of CN1 connection of the S signal

REFRIGERATION CYCLE DIAGRAM



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Fig. 7 — Refrigerant Cycle Diagram

REFRIGERANT LINES

General refrigerant line sizing:

1. The outdoor units are shipped with a full charge of R410A refrigerant. All charges, line sizing, and capacities are based on runs of 25 ft. (7.6 m). For runs over 25 ft. (7.6 m), consult the product data.
2. Minimum refrigerant line length between the indoor and outdoor units is 10 ft. (3 m).
3. Refrigerant lines should not be buried in the ground. If it is necessary to bury the lines, not more than 36 in. (914 mm) should be buried. Provide a minimum 6 in. (152 mm) vertical rise to the service valves to prevent refrigerant migration.
4. Both lines must be insulated. Use a minimum of 1/2 in. (12.7 mm) thick insulation. Closed-cell insulation is recommended in all long-line applications.
5. Special consideration should be given to isolating interconnecting tubing from the building structure. Isolate the tubing so that vibration or noise is not transmitted into the structure.
6. For piping runs greater than 25 ft. (7.6 m), add refrigerant up to the allowable length as specified in the product data.

SYSTEM EVACUATION AND CHARGING



CAUTION

UNIT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Never use the system compressor as a vacuum pump.

Refrigerant tubes and indoor coil should be evacuated using the recommended deep vacuum method of 500 microns. The alternate triple evacuation method may be used if the following procedure is followed. Always break a vacuum with dry nitrogen.

System Vacuum and Charge

Using Vacuum Pump

1. Completely tighten the flare nuts (A, B, C, D, E). Fully open all circuits service valves. Connect the manifold gage charge hose to the charge port of the low side Master service valve to evacuate all circuits at the same time (see Fig. 8).
2. Connect charge hose to vacuum pump.
3. Fully open the low side of manifold gage (see Fig. 9).
4. Start vacuum pump.
5. Evacuate using the triple evacuation method.
6. After evacuation is complete, fully close the low side of manifold gage and stop the vacuum pump operation.
7. The factory charge contained in the outdoor unit is good for up to 25 ft. (8 m) of line length.
8. Disconnect charge hose from charge connection of the low side service valve.
9. Fully open service valves B and A.
10. Securely tighten caps of service valves.

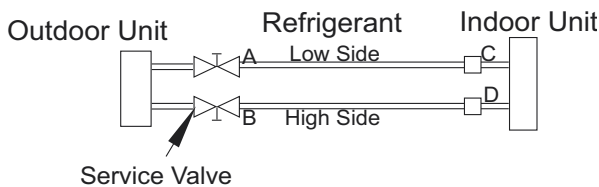


Fig. 8 — Service Valve

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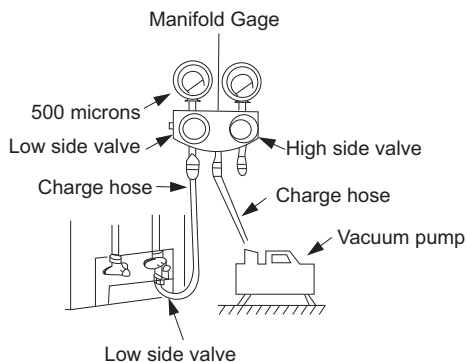


Fig. 9 — Manifold

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Deep Vacuum Method

The deep vacuum method requires a vacuum pump capable of pulling a vacuum of 500 microns and a vacuum gage capable of accurately measuring this vacuum depth. The deep vacuum method is the most positive way of ensuring a system is free of air and liquid water (see Fig. 10).

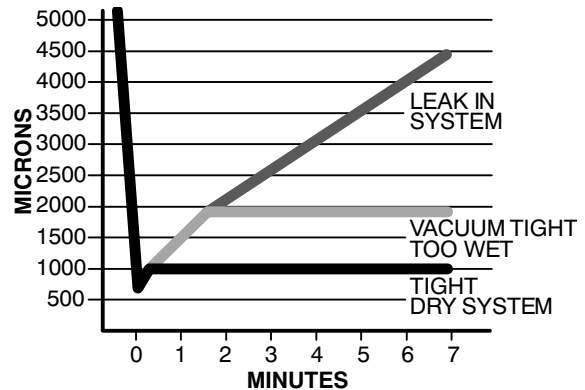


Fig. 10 — Deep Vacuum Method

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Triple Evacuation Method

The triple evacuation method should be used. Refer to Fig. 11 and proceed as follows:

1. Pump the system down to 1500 microns and allow the pump to continue operating for an additional 15 minutes.
2. Close the service valves and shut off the vacuum pump.
3. Connect a dry nitrogen cylinder and regulator to the system and break vacuum until the system reaches 2 psig.
4. Close the service valve and allow the system to stand for one hour. During this time, the dry nitrogen can diffuse throughout the system absorbing moisture.
5. Pump the system down to 1000 microns.
6. Break the vacuum with dry nitrogen (2 psig).
7. Pump the system down to 500 microns.
8. Perform the hold test for 30 minutes.

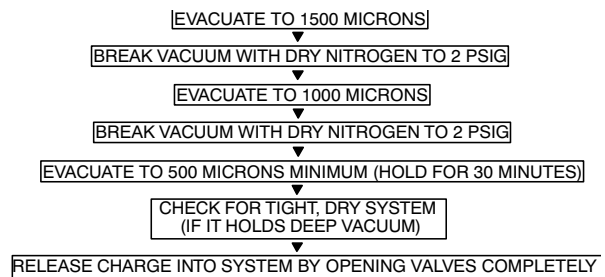


Fig. 11 — Triple Evacuation Method

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Final Tubing Check

IMPORTANT: Check to be certain factory tubing on both indoor and outdoor unit has not shifted during shipment. Ensure tubes are not rubbing against each other or any sheet metal. Pay close attention to feeder tubes, making sure wire ties on feeder tubes are secure and tight.

Main Protection
Fan Speed is Out of Control

When the indoor fan speed remains too low (lower than 300RPM) for 50 seconds, the indoor fan will shut off and restarts 30 seconds later, if protection occurred three times when the fan motor restarts continuously, the unit stops and the LED displays the failure.

When the outdoor fan speed remains too low (lower than 100 RPM) or too high (higher than 1500 RPM) for 60 seconds, the unit stops and the LED displays the failure. The malfunction is cleared 30 seconds later.

Indoor Fan Delayed Open Function

When the unit starts, the louver becomes active immediately and the indoor fan starts seven seconds later. If the unit runs in the HEATING mode, the indoor fan will be controlled by the anti-cold blow function.

Zero Crossing Detection Error Protection

If the AC detects that the time interval is not correct for a continuous period of 240 seconds, the unit stops and the LED displays the failure code. The correct zero crossing signal time interval should be between 6-13ms.

Sensor Protection at Open Circuit and Breaking Disconnection

When there is only one malfunctioning temperature sensor, the air conditioner keeps working yet displays the error code permitting emergency use. When there is more than one malfunctioning temperature sensor, the air conditioner stops working.

Evaporator low temperature T2 protection

- $T2 < 32^{\circ}\text{F}$ (0°C), the compressor stops and restarts when $T2 \geq 41^{\circ}\text{F}$ (5°C).
- 32°F (0°C) $\leq T2 < 39.2^{\circ}\text{F}$ (4°C), the compressor frequency is limited and decreased to the lower level.
- 39.2°F (4°C) $\leq T2 \leq 44.6^{\circ}\text{F}$ (7°C), the compressor maintains the current frequency
- $T2 > 44.6^{\circ}\text{F}$ (7°C), the compressor frequency is not limited.

Operation Modes and Functions
FAN Mode

1. Outdoor fan and compressor stop
2. Temperature setting function is disabled, and no setting temperature appears
3. Indoor fan can be set to high/med/low/auto
4. The louver operates the same in the COOLING mode
5. AUTO fan

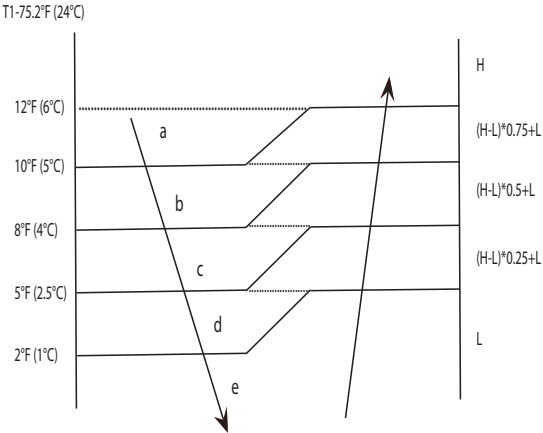


Fig. 12 —Auto Fan

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COOLING Mode

Indoor Fan Running Rules

In the COOLING mode, the indoor fan runs constantly and either speed (high, medium, low or auto) can be selected. When the setting temperature is reached, if the compressor stops running, the indoor fan motor runs at the minimum or setting speed. The indoor fan is controlled by the rules shown in Fig. 13.

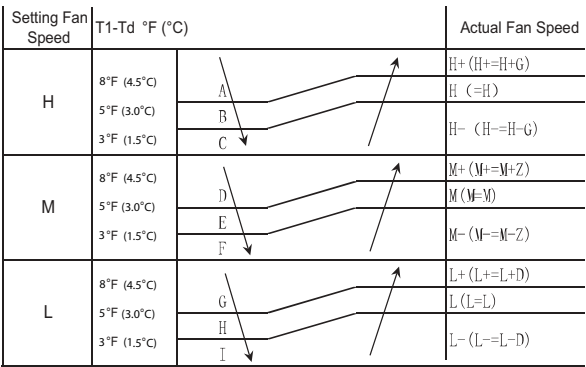
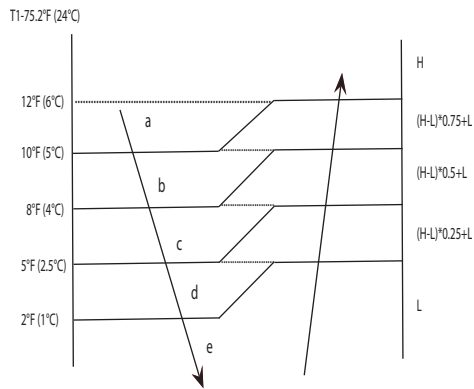


Fig. 13 — Indoor Fan Running Rules

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The **AUTO** fan is controlled by the rules shown in Fig. 14.



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Fig. 14 — Indoor Fan Running Rules

Evaporator Temperature Protection

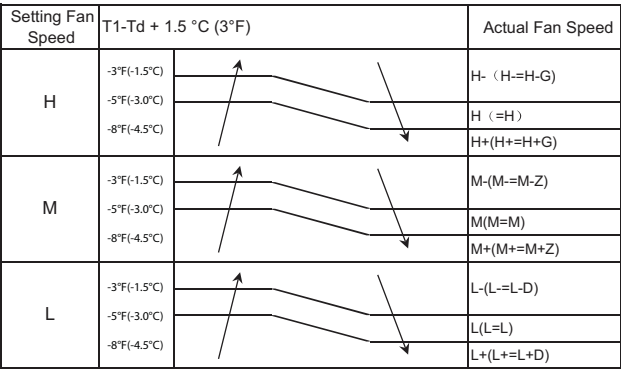
When the evaporator temperature is lower than the setting value, the compressor stops.

HEATING Mode

Indoor Fan Running Rules

When the compressor is on, the indoor fan can be set to high/med/low/ auto/mute. When the indoor unit coil temperature is low, the anti-cold air function starts and the indoor fan motor runs at a low speed and the speed can not be changed. When the temperature is lower than the setting value, the indoor fan motor stops.

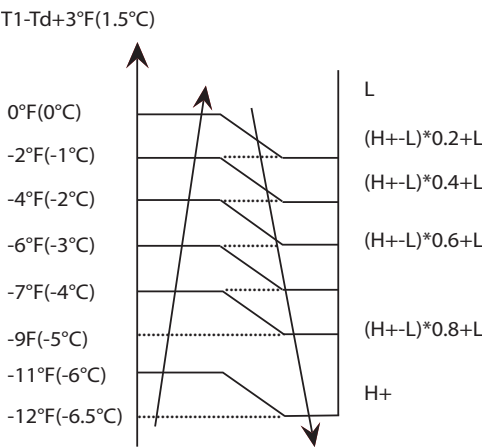
When the indoor temp reaches the setting temperature, the compressor stops and the indoor fan motor runs at the minimum or setting speed (the anti-cold air function is valid). The indoor fan is controlled by the rules shown in Fig. 15.



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Fig. 15 — Indoor Fan Running Rules

Auto Fan Action in HEATING Mode



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Fig. 16 — Auto Fan Action in HEATING Mode

DEFROST Mode

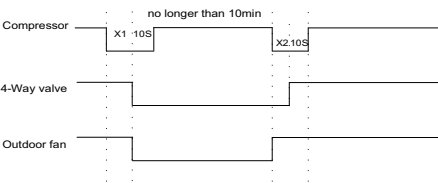
If any one of the following items is satisfied, the unit enters the **DEFROST** mode. After the compressor starts and runs, mark the minimum value of T3 from the 10th minute to the 15th minute as T30.

1. If the compressor runs for 29 minutes and $T3 < TCDI1$, $T3 + T30SUBT3ONE \leq T30$.
2. If the compressor runs for 35 minutes and $T3 < TCDI2$, $T3 + T30SUBT3TWO \leq T30$.
3. If the compressor runs for 29 minutes. and $T3 < TCDI3$ for three minutes.
4. If the compressor runs for 120 minutes and $T3 < 5^\circ\text{F} (-15^\circ\text{C})$.

Defrost End

If any one of the following items is satisfied, the **DEFROST** mode ends and the machine reverts to the normal **HEATING** mode.

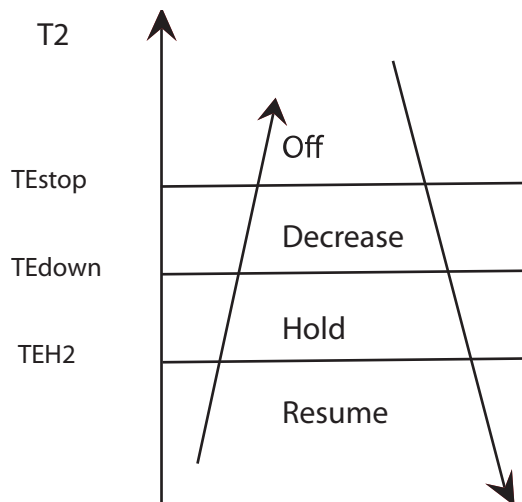
- T3 increases to a point higher than TCDE1.
- T3 maintains a point higher than TCDE2 for 80 seconds.
- Unit runs for 10 minutes in **DEFROST** mode.



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Fig. 17 — Defrost Action

Evaporator Coil Temperature Protection



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Fig. 18 — Evaporator Coil Temperature Protection

NOTE:

- **Off** – Compressor stops
- **Decrease** – Decrease running frequency to lower level
- **Hold** – Maintain the current frequency
- **Resume** – No frequency limit

When the evaporator temperature is higher than the setting protection value, the compressor stops.

Auto-Mode

This mode can be chosen with the remote control and the setting temperature can be changed between 62°F (17°C)~86°F (30°C).

In **AUTO** mode, the unit chooses the **COOLING**, **HEATING** or **FAN-ONLY** mode according to ΔT ($\Delta T = T1 - Ts$).

Table 4 — Auto Mode

$\Delta T = T1 - Ts$	Running Mode
$\Delta T > 4^{\circ}\text{F}$ (2°C)	Cooling
-4 (-2°C) $\leq \Delta T \leq 2^{\circ}\text{C}$ (4°F)	Fan-Only
$\Delta T < -2^{\circ}\text{C}$ (4°F)	Heating

The indoor fan runs under **AUTO** fan in the relevant mode. The louver operates the same as in the relevant mode. If the unit switches between the **HEATING** and **COOLING** modes, the compressor stops for 15 minutes and then chooses the mode according to $T1 - Ts$. If the setting temperature is modified, the unit chooses the running function again.

DRYING Mode

Indoor Fan Speed is Fixed

Indoor fan speed is fixed at breeze and can not be changed. The louver angle is the same as in the **COOLING** mode.

Low Indoor Room Temperature Protection

In the **DRYING** mode, if the room temperature is lower than 50°F (10°C), the compressor stops and will not resume until the room temperature exceeds 53.6°F (12°C).

Evaporator Anti-Freezing Protection

The evaporator anti-freezing protection condenser high temperature protection and outdoor unit frequency limit are active and both are the same when the unit is in the **COOLING** mode.

Timer Function

Timing range is 24 hours

- **Timer on:** The unit turns on automatically when reaching the setting time.
- **Timer off:** The unit turns off automatically when reaching the setting time.
- **Timer on/off:** The unit turns on automatically when reaching the setting “on” time, and then turns off automatically when reaching the setting “off” time.
- **Timer off/on:** The unit turns off automatically when reaching the setting “off” time, and then turns on automatically when reaching the setting “on” time.

The timer function will not change the unit’s current operation mode.

Suppose the unit is off now; it will not start up first after setting the “timer off” function. When the setting time is reached, the timer **LED** turns off and the unit’s running mode is the same. The setting time is relative time. The unit exits the timer function if it experiences a malfunction.

Sleep Function

The **SLEEP** function is available in the **COOLING**, **HEATING** or **AUTO** mode.

The Operation process in the **SLEEP** mode is as follows:

- **When cooling, the setpoint rises 2°F (1°C) (to a maximum of 86°F (30°C)) every one hour. Two (2) hours later, the room temperature stops rising and the indoor fan remains at a low speed.**
- **When heating, setpoint decreases 2°F (1°C) (to a minimum of 62°F (17°C)) every one hour. Two (2) hours later, the setpoint stops decreasing and the indoor fan is set at the low speed.**

(**Anti-Cold** wind function has the priority).

Operation time in the **SLEEP** mode is seven (7) hours. After seven (7) hours, the unit exits the **SLEEP** mode and powers off.

FORCED OPERATION Function

When the unit is off, press **TOUCH** to engage the **FORCED AUTO** mode. Press **TOUCH** again within five seconds to engage the **FORCED COOLING** mode. In the **FORCED AUTO**, **FORCED COOLING** or any other operation mode, press **TOUCH** to turn off the unit.

FORCED OPERATION Mode

In the **FORCED OPERATION** mode, all the general protections and remote control are available.

Operation Rules

FORCED COOLING Mode

The compressor runs at the F2 frequency and the indoor fan runs in the **BREEZE** mode. After running for 30 minutes, the machine enters **AUTO** mode at the 75.2°F (24°C) setting temperature.

FORCED AUTO Mode:

The **FORCED AUTO** mode is the same as the normal **AUTO** mode with a 75.2°F (24°C) setting temperature.

AUTO–RESTART Function

The indoor unit is equipped with an **AUTO–RESTART** function, which is carried out through an auto–restart module. In the event of a sudden power failure, the module memorizes the setting conditions prior to the power failure. The unit resumes the previous operation setting (not including the **SWING** function) automatically three (3) minutes after the power returns.

If the memorization condition is the **FORCED COOLING** mode, the unit runs in the **COOLING** mode for 30 minutes and switches to the **AUTO** mode at the 75°F (24°C) setpoint. If the unit is off before the power turns off and the unit is required to start up, the compressor delays the start up for 1 minute before powering on. In other instances, the compressor waits three (3) minutes before restarting.

FOLLOW ME

If the indoor PCB receives the signal, which results from pressing **FOLLOW ME** on the remote controller or wired remote controller, the buzzer will emit a sound and this indicates the **FOLLOW ME** function is initiated. However, when the indoor PCB receives the sent from the remote controller every three minutes, the buzzer will not respond. When the unit is running with the **FOLLOW ME** function, the PCB will control the unit according to the temperature from the **FOLLOW ME** signal, and the temperature collection function of the room temperature sensor will be shielded.

When the **FOLLOW ME** function is available, the PCB controls the unit according to the room temperature from the remote controller and the setting temperature. The PCB will change the mode based on information from the remote controller signal, but it will not be affected by the setting temperature.

If the unit is running under the **FOLLOW ME** function and the PCB does not receive a signal from the remote controller for seven minutes or after pressing **FOLLOW ME** again, the **FOLLOW ME** function will turn off automatically, and the temperature will control the unit according to the room temperature detected from its own room temperature sensor and setting temperature.

Refrigerant Leak Detection

With this new technology, the display area displays “EC” when the outdoor unit detects a refrigerant leak. This function is only active in the **COOLING** mode. This function helps to prevent the compressor from being damaged by a refrigerant leak or a compressor overload.

- **Open Condition:** When the compressor is running, the value of the T2 coil temperature sensor has no to very little change.

Louver Position Memory Function

When starting the unit again after powering down, the louver returns to the angle originally set by the user, however the precondition is that the angle must be within the allowable range. If the angle exceeds the allowable range, it memorizes the maximum angle of the louver. During operation, if the power fails or if the end user shuts down the unit while in the **TURBO** mode, the louver returns to the default angle.

46°F (8°C) Heating

When the compressor is running, the indoor fan motor runs without the **ANTI-COLD** air function. When the compressor is off, the indoor fan motor is off.

Silence Operation

Press **SILENCE** on the remote control to initiate the **SILENCE** function. When the **SILENCE** function is activated, the compressor running frequency remains lower than **F2** and the indoor unit emits a faint breeze, which reduces the noise to the lowest level and creates a quiet and comfortable space for the user.

Drain Pump Control

Use the water–level switch to control the action of drain pump.

NOTE: Main action under different condition: every five seconds the system will check the water level one time.

1. When the unit operates under **COOLING** mode (including auto cooling), dehumidifying, and forced cooling mode, the pump starts running immediately and continuously until cooling stops.
2. Once the water level increases to the control point, the LED alarm sounds, the drain pump opens and continues checking the water level. If the water level drops and the LED no longer alarms (drain pump delays for one minute) the system will operate under the last mode set. Otherwise, the entire system stops operating (including the pump) and the LED continues to alarm after three minutes.

Point Check Function

Press **LED DISPLAY** or **AIR DIRECTION** to check the next or front item's information. When the air conditioner enters the information enquiry status, it displays the code name in two seconds (see Table 5).

Table 5 — Information Enquiry

ENQUIRY INFORMATION	DISPLAYING CODE	MEANING
T1	T1	T1 temp.
T2	T2	T2 temp.
T3	T3	T3 temp.
T4	T4	T4 temp.
T2B	Tb	T2B temp.
T5	T5	T5 temp.
TH	TH	TH temp.
Targeted Frequency	FT	Targeted Frequency
Actual Frequency	Fr	Actual Frequency
Indoor fan speed	IF	Indoor fan speed
Outdoor fan speed	OF	Outdoor fan speed
EXV opening angle	LA	EXV opening angle
Compressor continuous running time	CT	Compressor continuous running time
Causes of compressor stop.	ST	Causes of compressor stop
Reserve	A0	
Reserve	A1	
Reserve	.b0	
Reserve	.b1	
Reserve	.b2	
Reserve	.b3	
Reserve	.b4	
Reserve	.b5	
Reserve	.b6	
Reserve	.d1	
Reserve	Ac	
Reserve	Uo	
Reserve	Td	

When the unit enters the information enquiry status, the code value appears for 25 seconds (see Table 6 on page 13).

Table 6 — Point Check

ENQUIRY INFORMATION	DISPLAY VALUE	MEANING	REMARK
T1,T2,T3,T4, T2B,TP,TH, Targeted Frequency, Actual Frequency	-1F,-1E,-1d,-1c,-1b,-1A	-25,-24,-23,-22,-21,-20	1. The displaying temperature is the actual value. 2. The temperature is °C no matter what kind of remote control is used. 3. T1,T2,T3,T4,T2B display range:-25~70, TP display range:-20~130. 4. Frequency display range: 0~159HZ. 5. If the actual value exceeds the range, it displays the maximum value or minimum value.
	-19—99	-19—99	
	A0,A1,...A9	100,101,...109	
	b0,b1,...b9	110,111,...119	
	c0,c1,...c9	120,121,...129	
	d0,d1,...d9	130,131,...139	
	E0,E1,...E9	140,141,...149	
	F0,F1,...F9	150,151,...159	
Indoor fan speed /Outdoor fan speed	0	OFF	
	1,2,3,4	Low speed, Medium speed, High speed, Turbo	For some big capacity motors
	14-FF	Actual fan speed = Display value turns to decimal value and then multiply by 10. The unit is RPM.	For some small capacity motors, the display value is from 14-F (hexadecimal), the corresponding fan speed range is from 200-2550 RPM.
EXV opening angle	0-FF	Actual EXV opening value = Display value turns to decimal value and then multiply by 2.	
Compressor continuous running time	0-FF	0-255 minutes	If the actual value exceeds the range, it displays the maximum value or minimum value.
Compressor stop causes	0-99	For a detailed meaning, please consult with an engineer.	Decimal display
Reserve	0-FF		

TROUBLESHOOTING

This section provides the required flow charts to troubleshoot problems that may arise.

NOTE: Information required in the diagnoses can be found either on the wiring diagrams or in the appendix.

Required Tools:

The following tools are needed when diagnosing the units:

- **Digital multimeter**
- **Screw drivers (Phillips and straight head)**
- **Needle-nose pliers**
- **Refrigeration gauges**

Recommended Steps

1. Refer to the diagnostic hierarchy charts below and determine the problem at hand.
2. Go to the chart listed in the diagnostic hierarchy and follow the steps in the chart for the selected problem.

For ease of service, systems are equipped with diagnostic code display LED's on both the indoor and outdoor units. The outdoor diagnostic display is on the outdoor unit board and is limited to very few errors. The indoor diagnostic display is a combination of flashing LED's on the display panel on the front of the unit. If possible always check the diagnostic codes displayed on the indoor unit first. The diagnostic codes for the indoor and outdoor units are listed in the appendix.

Safety

Electricity power is still kept in capacitors even the power supply is shut off. Do not forget to discharge the electricity power in capacitor.

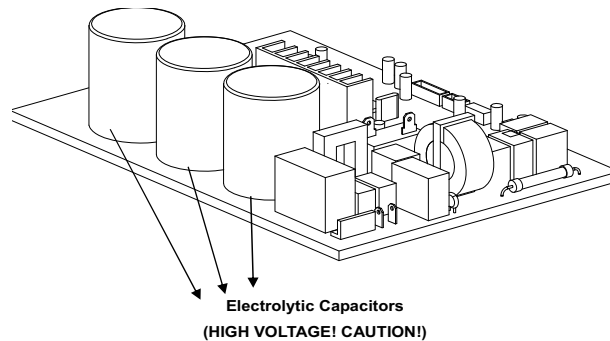


Fig. 19 — Capacitors

A220989

For other models, connect discharge resistance (approx. 100Ω 40W) or soldering iron (plug) between +, - terminals of the electrolytic capacitor on the contrary side of the outdoor PCB.

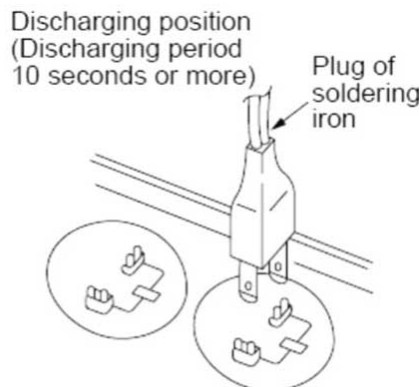


Fig. 20 — Discharging Position

A220990

NOTE: Figure 20 is for reference only.

Indoor Unit Diagnostic Guide

Table 7 — Indoor Unit Error Display

OPERATION LAMP	TIMER LAMP	DISPLAY	ERROR INFORMATION
1 time	OFF	EH00/EH0A	Indoor unit EEPROM parameter error
2 times	OFF	EL01	Indoor / outdoor unit communication error
4 times	OFF	EH03	The indoor fan speed is operating outside of the normal range (for some models)
6 times	OFF	EH60	Indoor room temperature sensor T1 is in open circuit or has short circuited
6 times	OFF	EH61	Evaporator coil temperature sensor T2 is in open circuit or has short circuited
8 times	OFF	EL0C	Refrigerant Leakage Detection (for some models)
13 times	OFF	EH0E	Water-level alarm malfunction
5 times	OFF	EC53	Outdoor room temperature sensor T4 is in open circuit or has short circuited
5 times	OFF	EC52	Condenser coil temperature sensor T3 is in open circuit or has short circuited
5 times	OFF	EC54	Compressor discharge temperature sensor TP is in open circuit or has short circuited
5 times	OFF	EC56	Evaporator coil outlet temperature sensor T2B is in open circuit or has short circuited (for free-match indoor units)
5 times	ON	EC51	Outdoor unit EEPROM parameter error
12 times	OFF	EC07	The outdoor fan speed is operating outside of the normal range (for some models)
7 times	FLASH	PC00	IPM malfunction or IGBT over-strong current protection
2 times	FLASH	PC01	Over voltage or over low voltage protection
3 times	FLASH	PC02	Top temperature protection of compressor or High temperature protection of IPM module
5 times	FLASH	PC04	Inverter compressor drive error
7 times	FLASH	PC03	High pressure protection or low pressure protection (for some models)
14 times	OFF	EC0d	Outdoor unit malfunction
1 time	ON	--	Indoor units mode conflict (match with multi outdoor unit)

NOTE: P3*

1. In the **HEATING** mode, if the outdoor temperature is lower than -13°F (-25°C) for 1 hour, the indoor unit displays a **P3** error code.
2. If the outdoor temperature is higher than -7.6°F (-22°C) for 10 minutes and the compressor stops for 1 hour or the outdoor temperature is higher than 23°F (-5°C) for 10 minutes, the unit returns to an operating mode.

* **Fault Symptom:** The display board shows a distorted code or a code that is not an error code found in the service manual nor a temperature reading.

DIAGNOSIS AND SOLUTION

EEPROM Parameter Error Diagnosis and Solution (E0/F4)

Error Code	E0/F4
Malfunction decision conditions	Indoor or outdoor PCB main chip does not receive feedback from the EEPROM chip
Supposed causes	• Installation mistake • PCB Faulty

Troubleshooting

Power OFF, then restart the unit 2 minutes later.

Yes

Replace the indoor/outdoor main PCB.

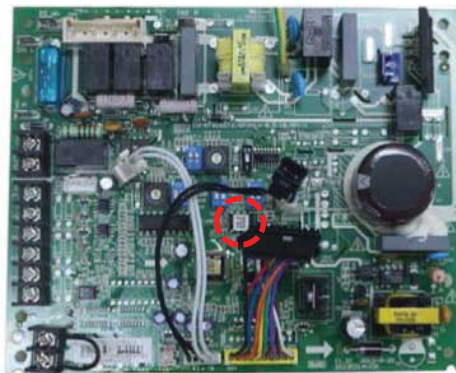


Fig. 21 — Indoor PCB

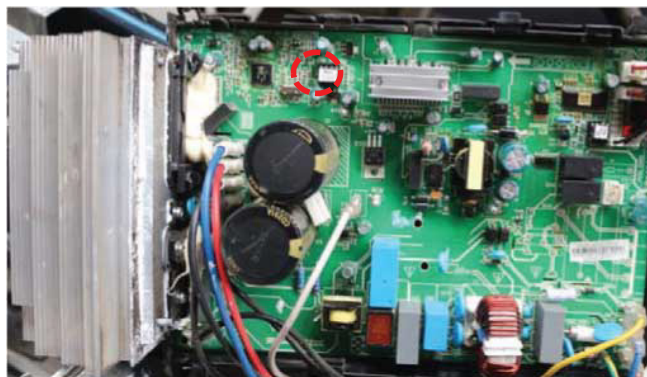


Fig. 22 — Outdoor PCB

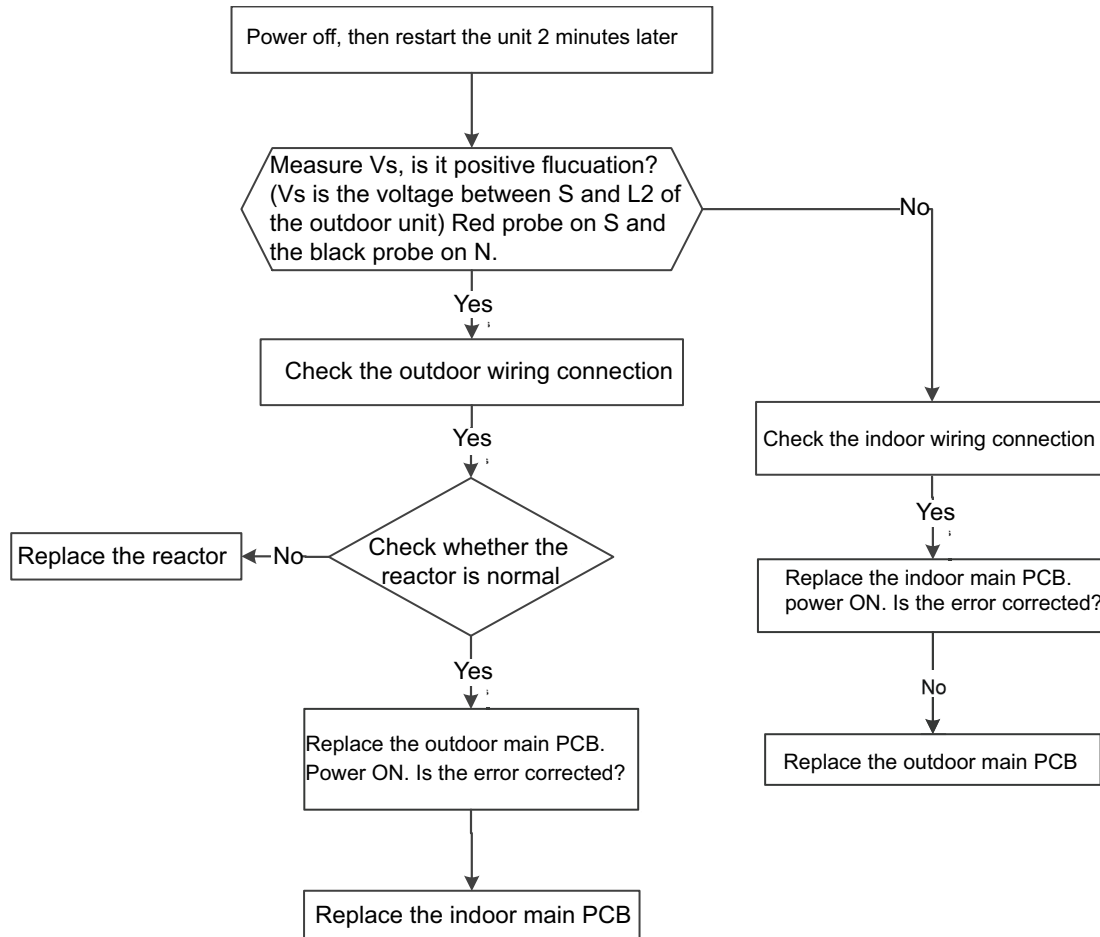
NOTE: Figures 21 and 22 are for reference only and may differ from your actual unit.

DIAGNOSIS AND SOLUTION (CONT.)

Indoor / outdoor unit's communication diagnosis and solution (E1)

Error Code	E1
Malfunction decision conditions	Indoor unit does not receive feedback from the outdoor unit for 110 seconds and this issue occurs four times continuously
Supposed causes	- Wiring mistake - Indoor or outdoor PCB Faulty

Troubleshooting



DIAGNOSIS AND SOLUTION (CONT.)



Fig. 23 —Test the DC voltage

NOTE: Use a multimeter to test the DC voltage between L2 port and S port of outdoor unit. The red probe of the multimeter connects with L2 port while the black pin is for S port. When the system is running normal, the voltage will move alternately between -50V to 50V. If the outdoor unit has a malfunction, the voltage moves alternately with positive value. While if the indoor unit has a malfunction, the voltage will be a fixed value.

DIAGNOSIS AND SOLUTION (CONT.)

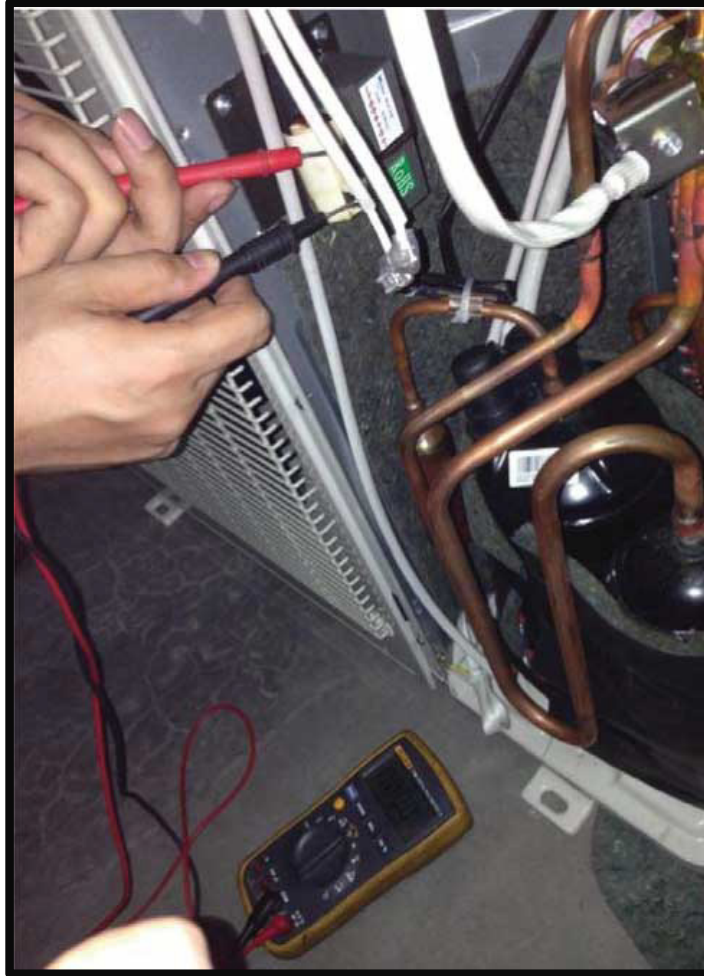


Fig. 24 — Test the Resistance

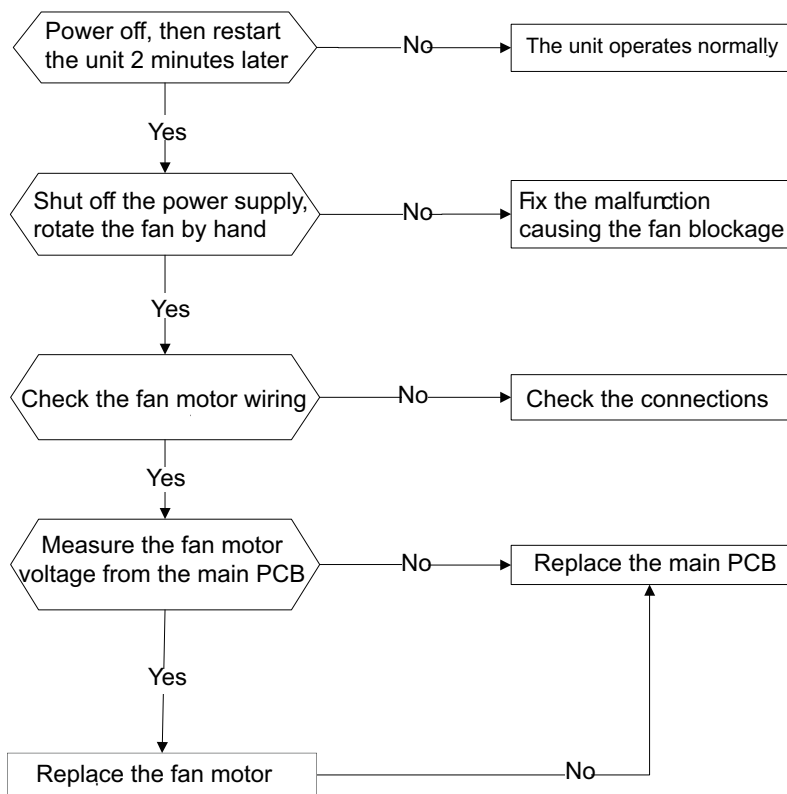
NOTE: Use a multimeter to test the resistance of the reactor with wiring disconnected. The normal value should be around zero ohm. Otherwise, the reactor is bad and needs to be replaced.

DIAGNOSIS AND SOLUTION (CONT.)

Fan speed is operating outside of the Normal Range (E3)

Error Code	E3
Malfunction decision conditions	When the indoor fan speed remains too low (300RPM) for a certain period of time, the unit stops and the LED displays the failure.
Supposed causes	• Wiring mistake • Fan assembly faulty • Fan motor faulty • PCB faulty

Troubleshooting



Index 1

Indoor DC fan motor (control chip is inside fan motor):

Power on and when the unit is in standby, measure the pin1-pin3, pin4-pin3 voltage in the fan motor connector. If the voltage value is not in the range shown in table 8, replace the PCB.

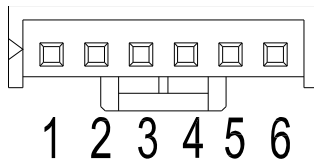


Fig. 25 — Indoor DC Fan Motor

DC motor voltage input and output

Table 8 — Signals

NO.	COLOR	SIGNAL	VOLTAGE
1	Red	Vs/Vm	200~380V
2	---	---	---
3	Black	GND	0V
4	White	Vcc	13.5~16.5V
5	Yellow	Vsp	0~6.5V
6	Blue	FG	13.5~16.5V

DIAGNOSIS AND SOLUTION (CONT.)

Open circuit or short circuit of temperature sensor diagnosis and solution (E4/E5/F1/F2/F3)

Error Code	E4/E5/F1/F2/F3
Malfunction decision conditions	If the sampling voltage is lower than 0.06V or higher than 4.94V, the LED displays the failure.
Supposed causes	• Wiring mistake • Sensor faulty • PCB faulty

Troubleshooting

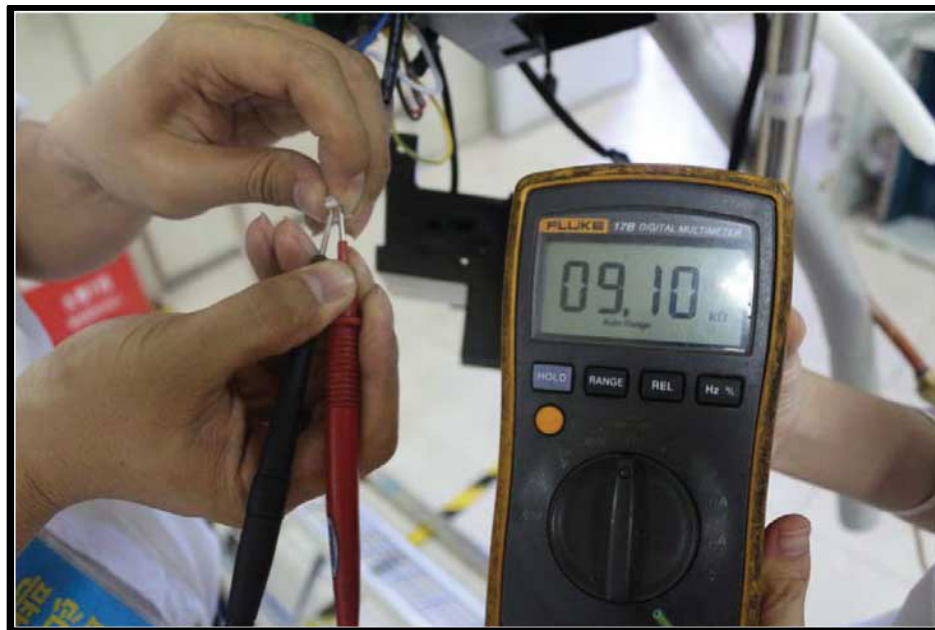
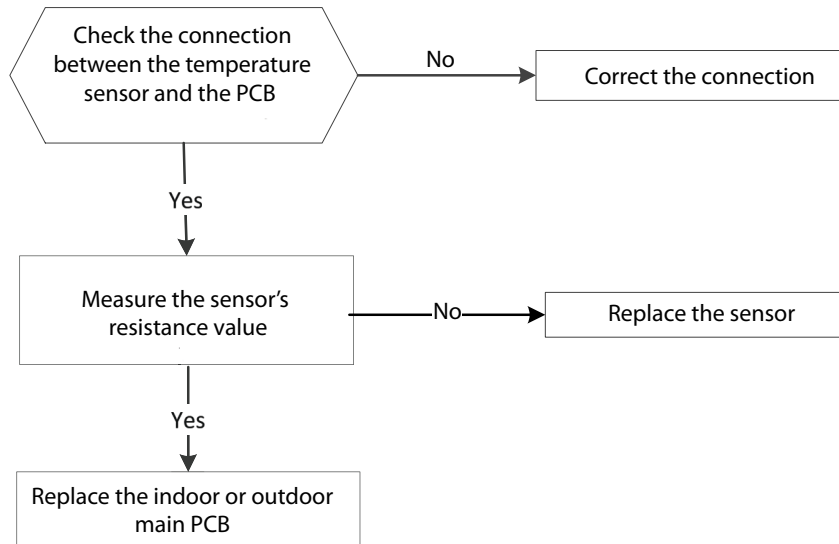


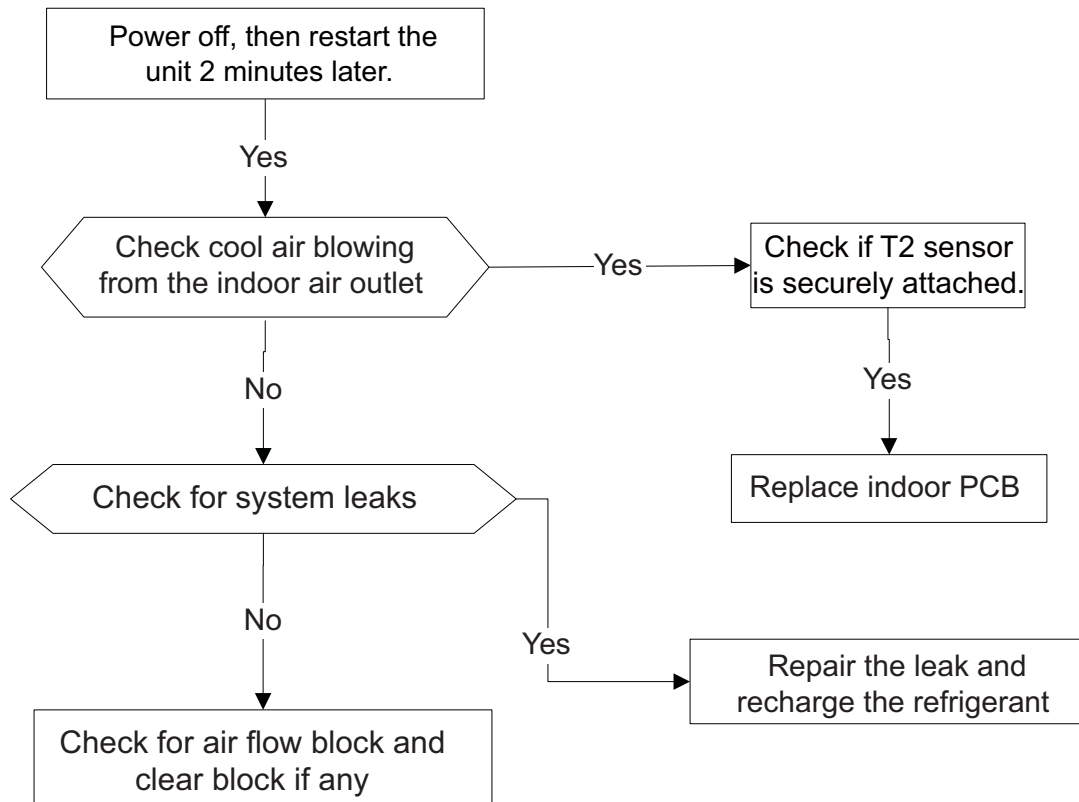
Fig. 26 — Temperature sensor diagnosis

DIAGNOSIS AND SOLUTION (CONT.)

Refrigerant Leakage Detection diagnosis and solution (EC)

Error Code	EC
Malfunction decision conditions	Define the evaporator coil temp.T2 of the compressor. It starts running in Tcool. At first, 5 minutes after the compressor starts up, if $T2 < T_{cool} - 35.6^{\circ}\text{F}$ ($T_{cool} - 2^{\circ}\text{C}$) does not run for 4 seconds and this situation occurs three times, the display area displays "EC" and the unit turns off.
Supposed causes	• Wiring mistake • T2 Sensor faulty • PCB faulty

Troubleshooting

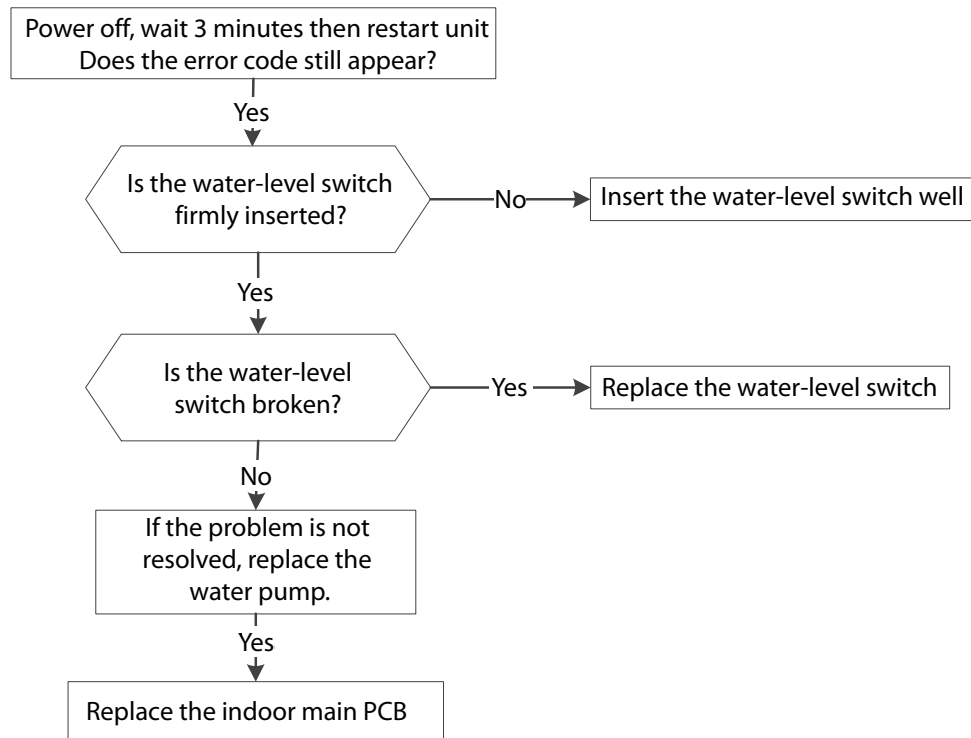


DIAGNOSIS AND SOLUTION (CONT.)

Water level alarm malfunction diagnosis and solution (EE)

Error Code	EE
Malfunction decision conditions	If the sampling voltage is not 5V, the LED will display the failure.
Supposed causes	• Wiring problem • Faulty water-level switch • Faulty water pump • Faulty indoor PCB

Troubleshooting

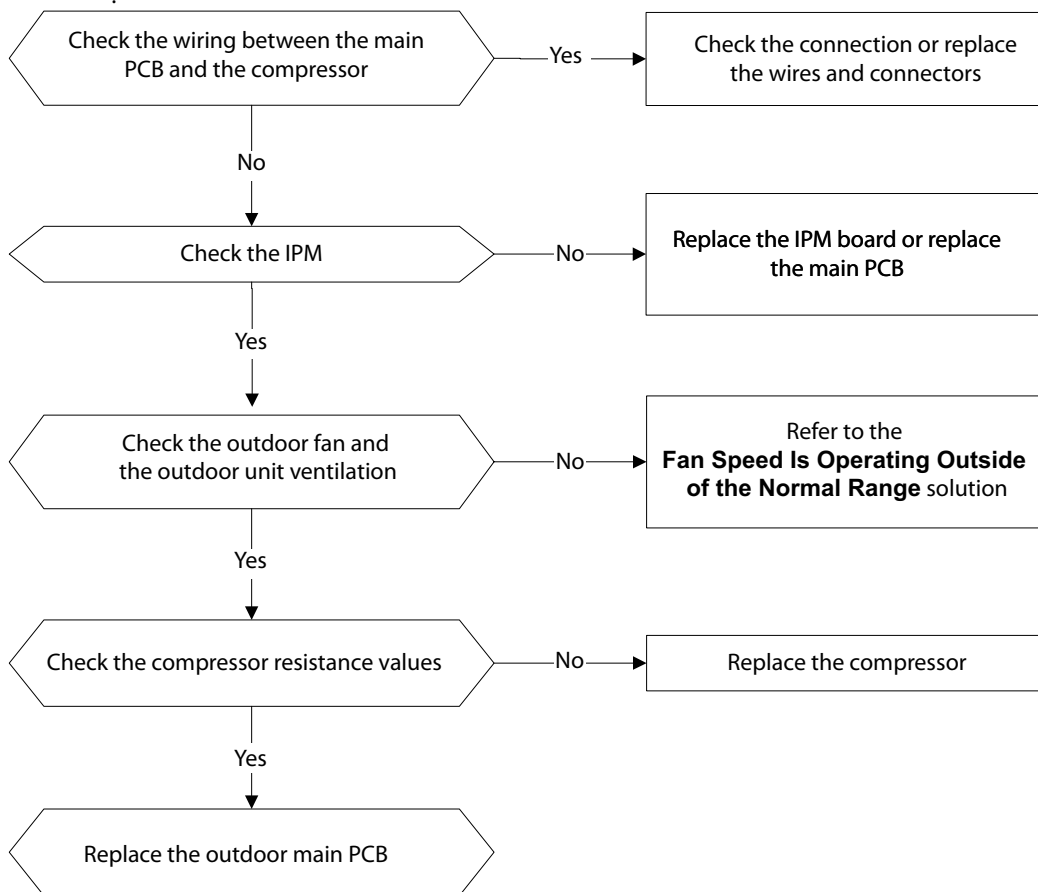


DIAGNOSIS AND SOLUTION (CONT.)

IPM malfunction or IGBT over-strong current protection diagnosis and solution (P0)

Error Code	P0
Malfunction decision conditions	When the voltage signal, that the IPM sends to the compressor drive chip is abnormal, the display LED displays P0 and the unit turns off.
Supposed causes	• Wiring mistake • IPM malfunction • Outdoor fan assembly faulty • Compressor malfunction • Outdoor PCB faulty

Troubleshooting



DIAGNOSIS AND SOLUTION (CONT.)

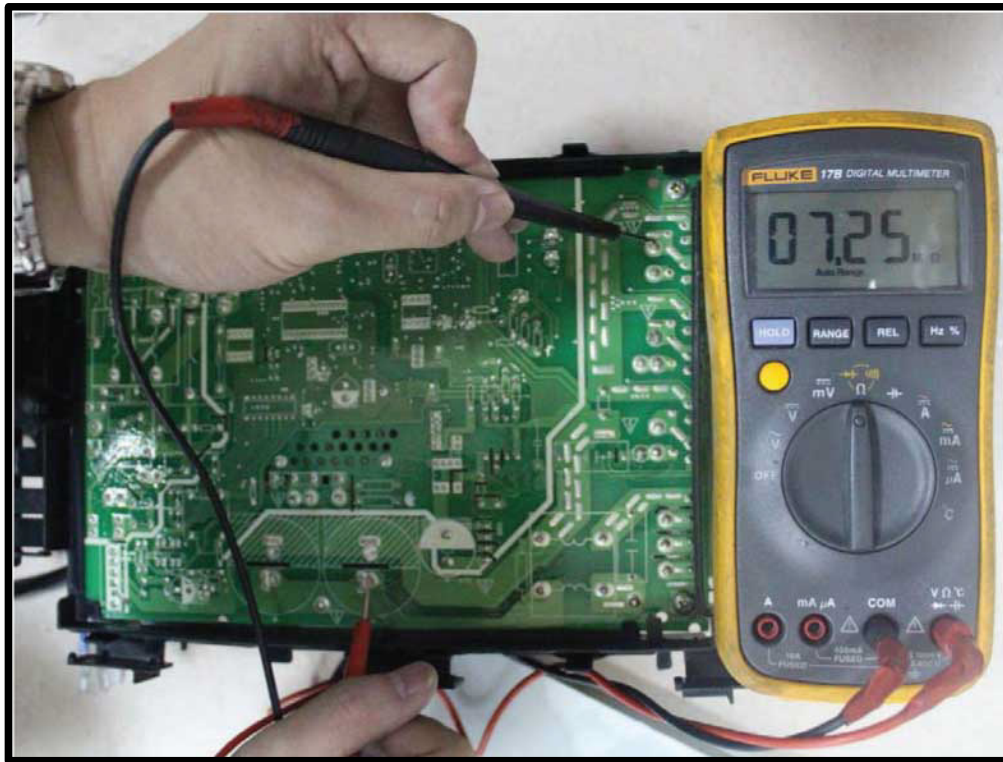


Fig. 27 — P-U

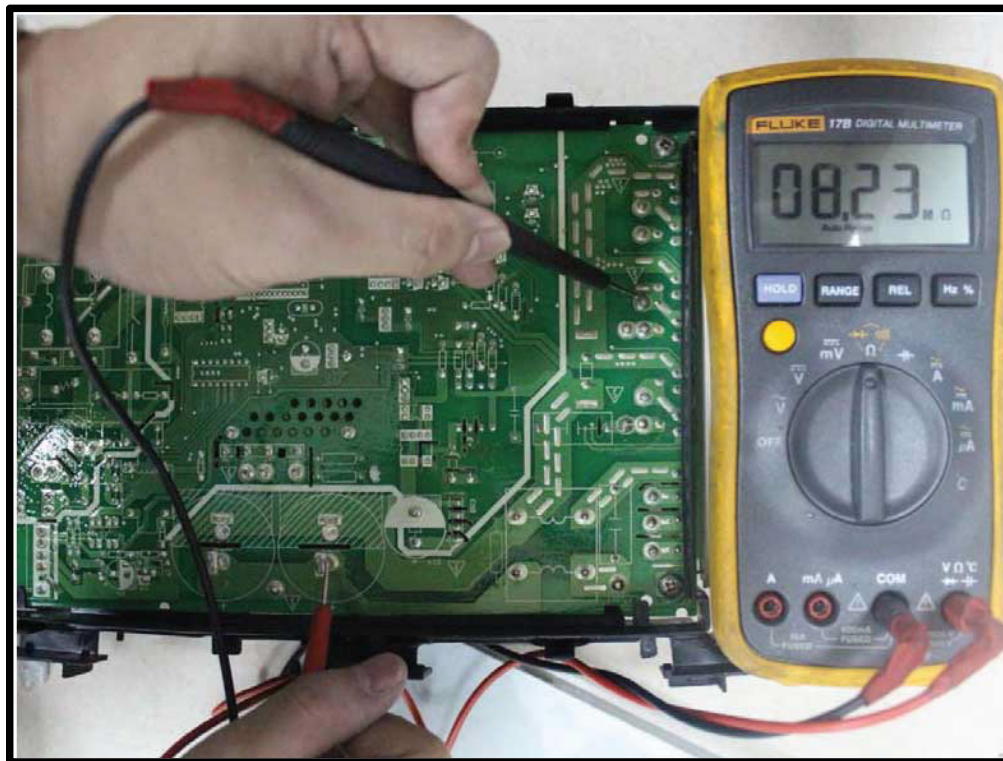


Fig. 28 — P-V

DIAGNOSIS AND SOLUTION (CONT.)

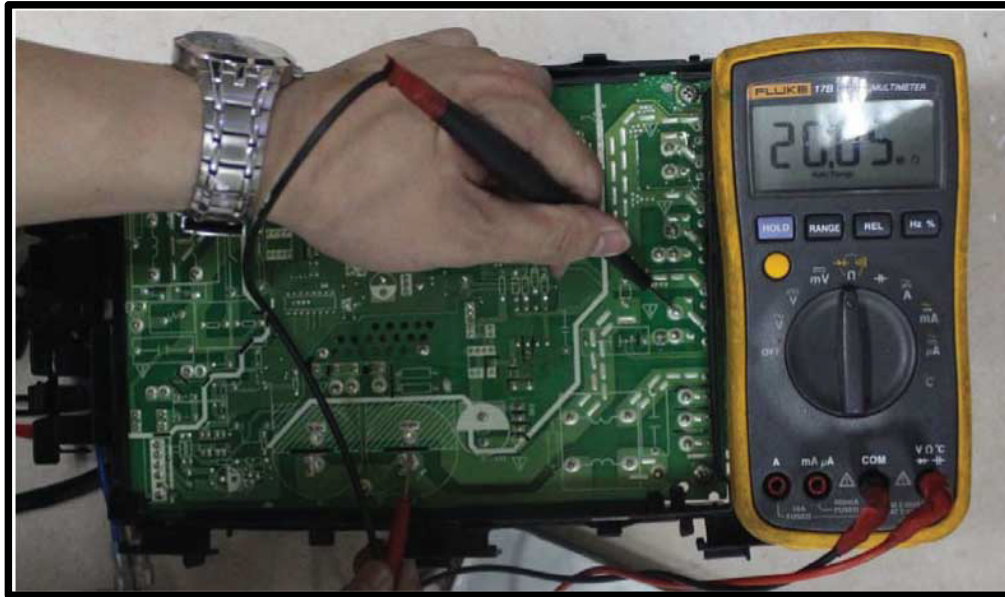


Fig. 29 — P-W

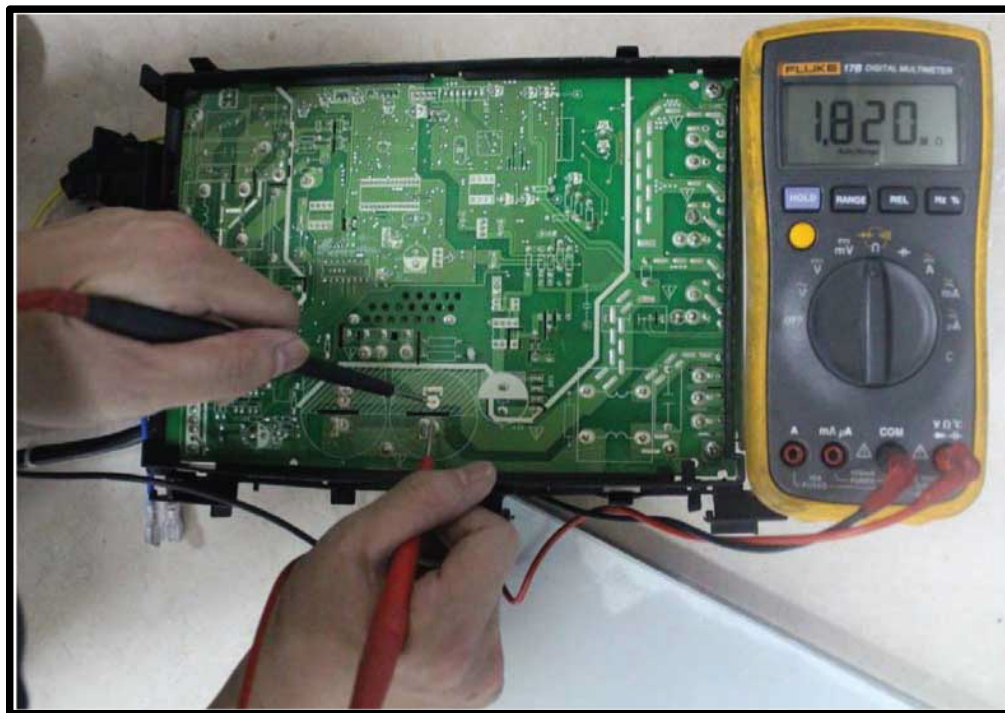


Fig. 30 — P-N

DIAGNOSIS AND SOLUTION (CONT.)

Over voltage or too low voltage protection diagnosis and solution (P1)

Error Code	P1
Malfunction decision conditions	An abnormal voltage rise or drop is detected by checking the specified voltage detection circuit.
Supposed causes	• Power supply functions • System leakage or block • PCB faulty

Troubleshooting

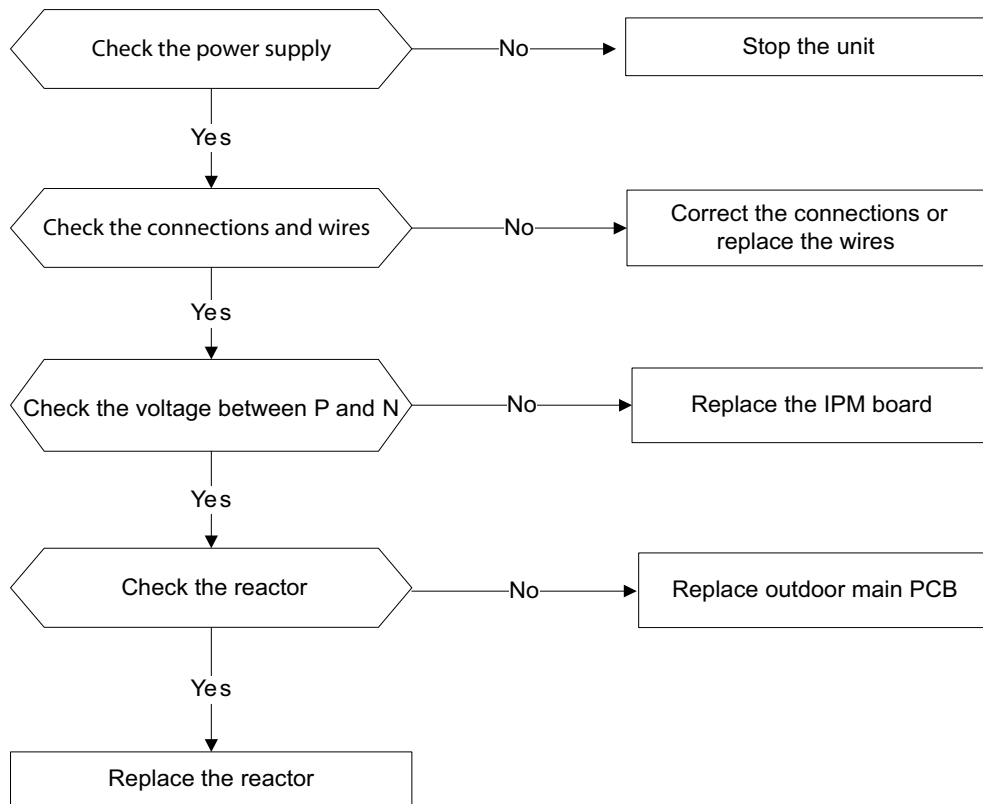


Fig. 31 — Measure the DC voltage

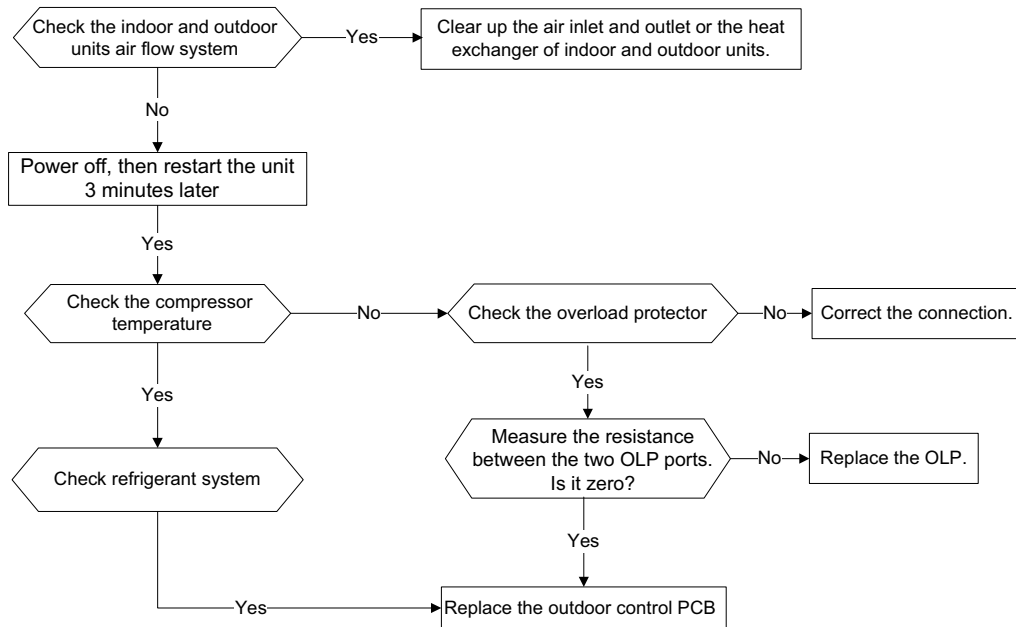
NOTE: Measure the DC voltage between ports P and N (the normal value should be around 310V).

DIAGNOSIS AND SOLUTION (CONT.)

High temperature protection of compressor top diagnosis and solution (P2)

Error Code	P2
Malfunction decision conditions	If the sampling voltage is not 5V, the LED displays the failure.
Supposed causes	• Power supply functions • System leakage or block • PCB faulty

Troubleshooting

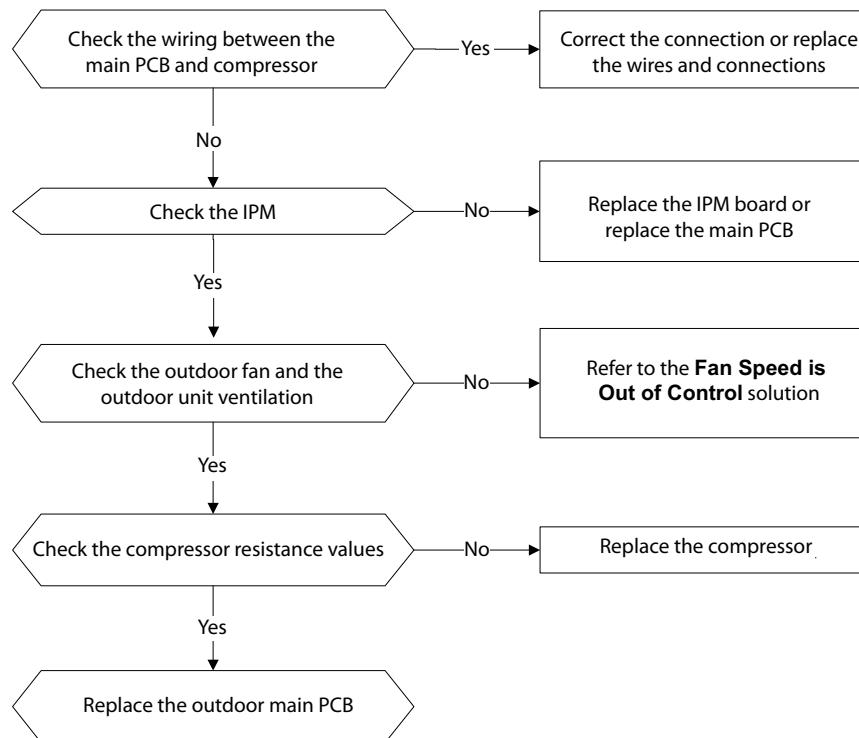


DIAGNOSIS AND SOLUTION (CONT.)

Inverter compressor drive error diagnosis and solution (P4)

Error Code	P4
Malfunction decision conditions	An abnormal inverter compressor drive is detected by a special detection circuit, including communication signal detection, voltage detection, compressor rotation speed signal detection, etc.
Supposed causes	• Wiring mistake • IPM malfunction • Outdoor fan assembly faulty • Compressor malfunction • Outdoor PCB faulty

Troubleshooting



Main Parts Check

Temperature Sensor Checking

Disconnect the temperature sensor from the PCB, measure the resistance value with a tester.

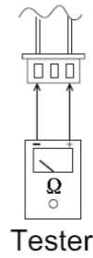


Fig. 32 —Tester

Temperature Sensors

Room temp.(T1) sensor,

Indoor coil temp.(T2) sensor,

Outdoor coil temp.(T3) sensor,

Outdoor ambient temp.(T4) sensor,

Compressor discharge temp.(T5) sensor.

Measure the resistance value of each winding by using the multi-meter.

APPENDICES

Appendix 1

Table 9 — Temperature Sensor Resistance Value Table for T1, T2, T3, T4 (°C--K)

°C	°F	K OHM	°C	°F	K OHM	°C	°F	K OHM	°C	° F	K OHM
-20	-4	115.266	20	68	12.6431	60	140	2.35774	100	212	0.62973
-19	-2	108.146	21	70	12.0561	61	142	2.27249	101	214	0.61148
-18	0	101.517	22	72	11.5	62	144	2.19073	102	216	0.59386
-17	1	96.3423	23	73	10.9731	63	145	2.11241	103	217	0.57683
-16	3	89.5865	24	75	10.4736	64	147	2.03732	104	219	0.56038
-15	5	84.219	25	77	10	65	149	1.96532	105	221	0.54448
-14	7	79.311	26	79	9.55074	66	151	1.89627	106	223	0.52912
-13	9	74.536	27	81	9.12445	67	153	1.83003	107	225	0.51426
-12	10	70.1698	28	82	8.71983	68	154	1.76647	108	226	0.49989
-11	12	66.0898	29	84	8.33566	69	156	1.70547	109	228	0.486
-10	14	62.2756	30	86	7.97078	70	158	1.64691	110	230	0.47256
-9	16	58.7079	31	88	7.62411	71	160	1.59068	111	232	0.45957
-8	18	56.3694	32	90	7.29464	72	162	1.53668	112	234	0.44699
-7	19	52.2438	33	91	6.98142	73	163	1.48481	113	235	0.43482
-6	21	49.3161	34	93	6.68355	74	165	1.43498	114	237	0.42304
-5	23	46.5725	35	95	6.40021	75	167	1.38703	115	239	0.41164
-4	25	44	36	97	6.13059	76	169	1.34105	116	241	0.4006
-3	27	41.5878	37	99	5.87359	77	171	1.29078	117	243	0.38991
-2	28	39.8239	38	100	5.62961	78	172	1.25423	118	244	0.37956
-1	30	37.1988	39	102	5.39689	79	174	1.2133	119	246	0.36954
0	32	35.2024	40	104	5.17519	80	176	1.17393	120	248	0.35982
1	34	33.3269	41	106	4.96392	81	178	1.13604	121	250	0.35042
2	36	31.5635	42	108	4.76253	82	180	1.09958	122	252	0.3413
3	37	29.9058	43	109	4.5705	83	181	1.06448	123	253	0.33246
4	39	28.3459	44	111	4.38736	84	183	1.03069	124	255	0.3239
5	41	26.8778	45	113	4.21263	85	185	0.99815	125	257	0.31559
6	43	25.4954	46	115	4.04589	86	187	0.96681	126	259	0.30754
7	45	24.1932	47	117	3.88673	87	189	0.93662	127	261	0.29974
8	46	22.5662	48	118	3.73476	88	190	0.90753	128	262	0.29216
9	48	21.8094	49	120	3.58962	89	192	0.8795	129	264	0.28482
10	50	20.7184	50	122	3.45097	90	194	0.85248	130	266	0.2777
11	52	19.6891	51	124	3.31847	91	196	0.82643	131	268	0.27078
12	54	18.7177	52	126	3.19183	92	198	0.80132	132	270	0.26408
13	55	17.8005	53	127	3.07075	93	199	0.77709	133	271	0.25757
14	57	16.9341	54	129	2.95896	94	201	0.75373	134	273	0.25125
15	59	16.1156	55	131	2.84421	95	203	0.73119	135	275	0.24512
16	61	15.3418	56	133	2.73823	96	205	0.70944	136	277	0.23916
17	63	14.6181	57	135	2.63682	97	207	0.68844	137	279	0.23338
18	64	13.918	58	136	2.53973	98	208	0.66818	138	280	0.22776
19	66	13.2631	59	138	2.44677	99	210	0.64862	139	282	0.22231

Appendix 2

Table 10 — Temperature Sensor Resistance Value Table for T5 (TP)(°C--K)

° C	° F	K OHM	° C	° F	K OHM	° C	° F	K OHM	° C	° F	K OHM
-20	-4	542.7	20	68	68.66	60	140	13.59	100	212	3.702
-19	-2	511.9	21	70	65.62	61	142	13.11	101	214	3.595
-18	0	483	22	72	62.73	62	144	12.65	102	216	3.492
-17	1	455.9	23	73	59.98	63	145	12.21	103	217	3.392
-16	3	430.5	24	75	57.37	64	147	11.79	104	219	3.296
-15	5	406.7	25	77	54.89	65	149	11.38	105	221	3.203
-14	7	384.3	26	79	52.53	66	151	10.99	106	223	3.113
-13	9	363.3	27	81	50.28	67	153	10.61	107	225	3.025
-12	10	343.6	28	82	48.14	68	154	10.25	108	226	2.941
-11	12	325.1	29	84	46.11	69	156	9.902	109	228	2.86
-10	14	307.7	30	86	44.17	70	158	9.569	110	230	2.781
-9	16	291.3	31	88	42.33	71	160	9.248	111	232	2.704
-8	18	275.9	32	90	40.57	72	162	8.94	112	234	2.63
-7	19	261.4	33	91	38.89	73	163	8.643	113	235	2.559
-6	21	247.8	34	93	37.3	74	165	8.358	114	237	2.489
-5	23	234.9	35	95	35.78	75	167	8.084	115	239	2.422
-4	25	222.8	36	97	34.32	76	169	7.82	116	241	2.357
-3	27	211.4	37	99	32.94	77	171	7.566	117	243	2.294
-2	28	200.7	38	100	31.62	78	172	7.321	118	244	2.233
-1	30	190.5	39	102	30.36	79	174	7.086	119	246	2.174
0	32	180.9	40	104	29.15	80	176	6.859	120	248	2.117
1	34	171.9	41	106	28	81	178	6.641	121	250	2.061
2	36	163.3	42	108	26.9	82	180	6.43	122	252	2.007
3	37	155.2	43	109	25.86	83	181	6.228	123	253	1.955
4	39	147.6	44	111	24.85	84	183	6.033	124	255	1.905
5	41	140.4	45	113	23.89	85	185	5.844	125	257	1.856
6	43	133.5	46	115	22.89	86	187	5.663	126	259	1.808
7	45	127.1	47	117	22.1	87	189	5.488	127	261	1.762
8	46	121	48	118	21.26	88	190	5.32	128	262	1.717
9	48	115.2	49	120	20.46	89	192	5.157	129	264	1.674
10	50	109.8	50	122	19.69	90	194	5	130	266	1.632
11	52	104.6	51	124	18.96	91	196	4.849			
12	54	99.69	52	126	18.26	92	198	4.703			
13	55	95.05	53	127	17.58	93	199	4.562			
14	57	90.66	54	129	16.94	94	201	4.426			
15	59	86.49	55	131	16.32	95	203	4.294			
16	61	82.54	56	133	15.73	96	205	4.167			
17	63	78.79	57	135	15.16	97	207	4.045			
18	64	75.24	58	136	14.62	98	208	3.927			
19	66	71.86	59	138	14.09	99	210	3.812			

Appendix 3

Table 11 — Celsius to Fahrenheit

° C	° F	° C	° F	° C	° F	° C	° F	° C	° F
-5	23	21	69.8	51	123.8	82	179.6	113	235.4
-4	24.8	22	71.6	52	125.6	83	181.4	114	237.2
-3	26.6	23	73.4	53	127.4	84	183.2	115	239
-2	28.4	24	75.2	54	129.2	85	185	116	240.8
-1	30.2	25	77	55	131	86	186.8	117	242.6
0	32	25.5	77.9	56	132.8	87	188.6	118	244.4
0.5	32.9	26	78.8	57	134.6	88	190.4	119	246.2
1	33.8	27	80.6	58	136.4	89	192.2	120	248
1.5	34.7	28	82.4	59	138.2	90	194	121	249.8
2	35.6	29	84.2	60	140	91	195.8	122	251.6
2.5	36.5	30	86	61	141.8	92	197.6	123	253.4
3	37.4	31	87.8	62	143.6	93	199.4	124	255.2
3.5	38.3	32	89.6	63	145.4	94	201.2	125	257
4	39.2	33	91.4	64	147.2	95	203	126	258.8
4.5	40.1	34	93.2	65	149	96	204.8	127	260.6
5	41	35	95	66	150.8	97	206.6	128	262.4
6	42.8	36	96.8	67	152.6	98	208.4	129	264.2
7	44.6	37	98.6	68	154.4	99	210.2	130	266
8	46.4	38	100.4	69	156.2	100	212	131	267.8
9	48.2	39	102.2	70	158	101	213.8	132	269.6
10	50	40	104	71	159.8	102	215.6	133	271.4
11	51.8	41	105.8	72	161.6	103	217.4	134	273.2
12	53.6	42	107.6	73	163.4	104	219.2	135	275
13	55.4	43	109.4	74	165.2	105	221	136	276.8
14	57.2	44	111.2	75	167	106	222.8	137	278.6
15	59	45	113	76	168.8	107	224.6	138	280.4
16	60.8	46	114.8	77	170.6	108	226.4	139	282.2
17	62.6	47	116.6	78	172.4	109	228.2	140	284
18	64.4	48	118.4	79	174.2	110	230	141	285.8
19	66.2	49	120.2	80	176	111	231.8	142	287.6
20	68	50	122	81	177.8	112	233.6	143	289.4

IPM Continuity Check

Turn off the power, let the large capacity electrolytic capacitors discharge completely, and dismount the IPM. Use a digital tester to measure the resistance between P and UVWN; UVW and N.

Table 12 — IPM Continuity Check

DIGITAL TESTER		NORMAL RESISTANCE VALUE	DIGITAL TESTER		NORMAL RESISTANCE VALUE
(+) Red	(-) Black	∞ (Several M W)	(+) Red	(-) Black	∞ (Several M W)
P	N		U	N	
	U		V		
	V		W		
	W		(+) Red		

Pressure on Service Port

Table 13 — Cooling Chart

°F °C	INDOOR TEMP.	OUTDOOR TEMP.				
		75 (23.89)	85 (29.44)	95 (35)	105 (40.56)	115 (46.11)
BAR	70	8.2	7.8	8.1	8.6	10.1
BAR	75	8.6	8.3	8.7	9.1	10.7
BAR	80	9.3	8.9	9.1	9.6	11.2
PSI	70	119	113	117	125	147
PSI	75	124	120	126	132	155
PSI	80	135	129	132	140	162
MPA	70	0.82	0.78	0.81	0.86	1.01
MPA	75	0.86	0.83	0.87	0.91	1.07
MPA	80	0.93	0.89	0.91	0.96	1.12

Heating Chart

Table 14 — Heating Chart

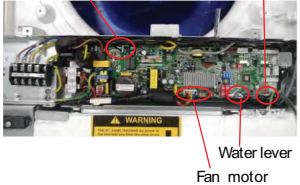
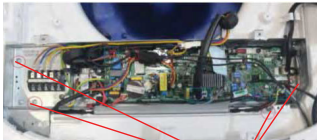
°F/°C	INDOOR TEMPERATURE	OUTDOOR TEMPERATURE				
		57 (13.89)	47 (8.33)	37 (2.78)	27 (-2.78)	17 (-8.33)
BAR	55	30.3	28.5	25.3	22.8	20.8
BAR	65	32.5	30.0	26.6	25.4	23.3
BAR	75	33.8	31.5	27.8	26.3	24.9
PSI	55	439	413	367	330	302
PSI	65	471	435	386	368	339
PSI	75	489	457	403	381	362
MPA	55	3.03	2.85	2.53	2.28	2.08
MPA	65	3.25	3.00	2.66	2.54	2.33
MPA	75	3.38	3.15	2.78	2.63	2.49

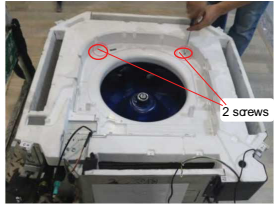
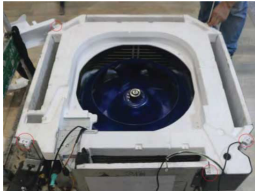
DISASSEMBLY INSTRUCTIONS

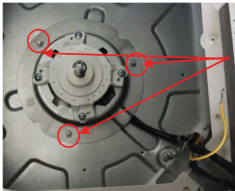
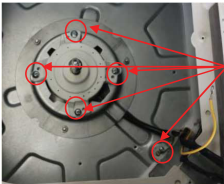
NOTE: This section is for reference, the photos may differ slightly from the actual unit.

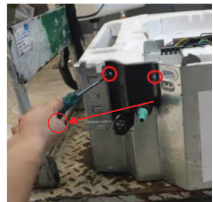
(Sheet 1 of 5)

NO.	PARTS NAME	PROCEDURES	REMARKS
	FRONT PANEL		
1	Remove the filter	1. Push one side of the grille clamp.	
		2. Remove one screw then push two grille clamps to remove the air inlet grille assembly.	 
		3. Turn over the air inlet grille assembly then pull up the filter. NOTE: The filter can be easily damaged, therefore be careful when removing it.	
2	DISPLAY BOARD	REMOVE THE DISPLAY BOARD	
		1. Open the grille. Repeat steps 1 and 2 found in section 1.	
		2. Disassemble the display board: • Open the install cover assembly (with display board) • Remove 1 screw of the display window board • Turn over the display board, push the switch to remove the display board.	

	PCB	REMOVE THE PCB	
		1. Open the grille. Repeat steps 1 and 2 found in section 1. There is no need to remove the panel. 2. Disassemble the electronic control box cover. 3. Remove the 2 screws.	 2 screws
3		4. Pull out all the connection wires attached to other parts. The PCB can now be replaced.	
		5. Remove the 2 main control board screws and the 2 ground wire screws.	 2 ground wire screws
	CONTROL BOX	REMOVE THE ELECTRONIC CONTROL BOX	
		1. Open the grille. Repeat steps 1 and 2 found in section 1. There is no need to take down the panel. 2. Remove the electronic control box cover. Repeat step 2 from section 3. 3. Remove all the plugs or connectors connected to the electronic control box.	
4		4. Remove the electronic control box. Remove the 3 screws of the electronic control box and the 1 ground wire.	 4 screws

5		REMOVE THE FAN WHEEL	
		1. Repeat the steps from section 4.	
		2. Remove the ventilation ring. Release the 2 screws to disassemble.	
		3. Remove the 2 screws securing the water collector.	
		4. Remove the 5 screws securing the water collector subassembly.	
		5. Turn over the water collector subassembly and remove the water level switch.	
		6. Remove the nut to disassemble the fan wheel.	
		7. Pull out the fan wheel.	

6	FAN MOTOR	REMOVE THE FAN MOTOR	
		1. Repeat steps found in section 5.	
		2. Remove the fan motor wire board.	
7	DRAINING PUMP	REMOVE THE DRAINING PUMP	
		1. Remove the panel. Repeat the steps from section 1.	
		2. Remove the 5 screws securing the external water pump box assembly.	
			
		3. Remove the water pump box assembly.	

8	EVAPORATOR	REMOVE THE EVAPORATOR	
		1. Remove the water collecting assembly. Repeat steps 1-4 from section 6.	
		2. Remove the evaporator's seal board (2 screws).	
		3. Remove the evaporator board (4 screws).	  

